

Viscereality: Bioresponsive Virtual Reality for Box Breathing and Altered-State Experience

George Fejer^{1,2,3*}, Till Holzapfel⁴, Taru Hirvonen,
Anestis Lalidis Mateo⁶, Johannes Blum⁵, Michael Gaebler²,
Bigna Lenggenhager^{7,8}

^{1*}University of Konstanz, Konstanz, Germany.

²Max Planck Institute for Human Cognitive and Brain Sciences,
Germany.

³ALIUS Research Network, Germany.

⁴University of Duisburg-Essen, Research Center Trustworthy Data
Science and Security, Duisburg, Germany.

⁵Pädagogische Hochschule Schaffhausen, Schaffhausen, Switzerland.

⁶Institute for Building Intelligence, Hochschule Bielefeld - University of
Applied Sciences and Arts, Bielefeld, Germany.

⁷AIR—Association for Independent Research, Zurich, Switzerland.

⁸Department of Psychology, University of Zurich, Zurich, Switzerland.

*Corresponding author(s). E-mail(s): george.fejer@aliusresearch.org;
Contributing authors: tillholzapfel@gmail.com; 2taruhi@gmail.com;
alalidis@hsbi.de; johannes.blum@phsh.ch; gaebler@mpib-berlin.mpg.de;
bigna.lenggenhager@gmail.com;

Abstract

Breath-based interactions in virtual reality typically map breathing onto discrete visual objects or physiological proxies. Viscereality instead uses a particle field to translate internal physiological state into an exteroceptive representation, mapping lung volume onto the surrounding space, so that inhalation expands the environment and exhalation contracts it. Unlike discrete objects, particle representations have no fixed shape or boundary, and carry information through the position, motion, colour, and size of individual units and through their interactions, such as synchronization with nearest neighbours. Here we investigate a VR system developed to augment breathing practices and modulate altered states of consciousness by making these dynamics more fluid and

synchronous during breath retention, and whether users are aware of this. In a preregistered within-subjects study, participants completed three ten-minute audio-guided box-breathing sessions in counterbalanced order: a high-symmetry VR condition in which particles transiently aligned into geometric coherence during breath holds, an asymmetric condition in which they remained irregular, and a dark-screen control. Subjective experience was assessed with an altered states of consciousness questionnaire and pictographic self-related measures. Relative to the dark-screen control, VR conditions improved breathing synchronization, expanded the spatial frame of self-reference, and produced altered-state profiles with strong perceptual, moderate pleasant, and weak distressing effects. The symmetry manipulation registered implicitly in participants' experience without supporting explicit identification of which session was symmetric. After a single ten-minute session, perceptual alterations and pleasant experiences were comparable to those reported for hyperventilation-based breathwork of longer duration, while distress was lower.

Keywords: biofeedback, breath-based interaction, breathwork, altered states of consciousness, particle systems

1 Introduction

1.1 Breath as an Affective Sensory Modality

Breathing is a unique and multifaceted sensory modality, as it carries both interoceptive and exteroceptive streams of information. One stream concerns internal sensory information about respiratory mechanics and blood-gas status, including signals related to lung inflation, airway and respiratory-muscle activity, and blood O_2/CO_2 balance (Chan et al. 2024; Engelen et al. 2023; Weng et al. 2021). Because inhaled air stems from the external environment, breathing also samples exteroceptive information about ambient conditions, including odours, airflow, and air temperature, through its coupling with orthonasal olfaction and thermo- and mechanosensory receptors in the upper airways (Mori and Sakano 2022a,b; Zhou et al. 2011; Sabnis et al. 2008; Kytikova et al. 2021). Breathing is also closely tied to affective experience. People sigh with relief after stress (Balban et al. 2023), hold their breath during freeze responses (Boiten et al. 1994), and hyperventilate during fear or panic (Jerath and Beveridge 2020; Kreibig 2010). Arousing negative emotions such as fear and anger are linked to shallower, more rapid breathing, whereas relaxation and contentment correspond to slower, deeper respiratory patterns (Boiten et al. 1994; Homma and Masaoka 2008; Jerath and Beveridge 2020). Unlike most visceral signals such as heartbeat, breathing is also amenable to voluntary control (Sarkar 2017), making this relationship bidirectional. Voluntarily changing respiratory patterns can modulate affective state (Jerath and Beveridge 2020; Ashhad et al. 2022), a principle that underlies breathwork as a therapeutic modality (Fincham et al. 2023; Balban et al. 2023; Siebieszuk et al. 2025). Breathwork techniques range from slow-paced protocols that target relaxation, stress regulation, and improvements in heart rate variability through sustained

cardiovascular-respiratory coupling (Balban et al. 2023; Ahmed et al. 2021; Siebieszuk et al. 2025), to high-ventilation protocols such as cyclic hyperventilation that can facilitate more pronounced alterations in affect and consciousness (Bahi et al. 2024a; Lewis-Healey et al. 2024; Havenith et al. 2025a), effects increasingly posited as therapeutic, for instance in facilitating emotional breakthroughs (Fincham et al. 2025, 2024). Breathwork is already established as a therapeutic modality for stress, anxiety, and depression (Fincham et al. 2023; Bentley et al. 2023; Banushi et al. 2023), and a growing number of virtual-reality applications also incorporate breathing for therapeutic and health-related purposes (Cortez-Vázquez et al. 2024; Pancini et al. 2025). Yet this literature has so far explored only a relatively limited portion of the broader design space that breathing affords as an interaction modality in VR.

1.2 Breath-Based Interactions in VR

Existing breath-based VR systems span meditation (Prpa et al. 2020; Shamekhi and Bickmore 2018), anxiety regulation (Bossenbroek et al. 2020), stress management through heart-rate-variability biofeedback (Blum et al. 2019; Rockstroh et al. 2019), and rehabilitation (Shi et al. 2023; Alhammad 2024). In the context of our framework, we distinguish between two motifs in the design space regarding the mapping of the user’s respiratory signal onto discrete visual objects, in contrast to environmental mappings. Object-oriented approaches typically place respiratory feedback in a discrete visual target. Third-person variants let the user watch an external entity respond to breathing, such as a cloud that approaches during inhalation and recedes during exhalation (Tinga et al. 2019). In first-person variants, the mapping is applied to a virtual body seen from the user’s own vantage point. The “embreathment” illusion demonstrated that synchronising a virtual avatar’s chest movements with the user’s breathing enhances body ownership and agency (Monti et al. 2020a), and mapping respiratory signals onto the expansion and contraction of a virtual avatar bubble viewed from a third-person perspective has been found to enhance agency and body awareness (Wald et al. 2023). What these object-oriented designs share is that respiratory feedback is concentrated in a single visual target, whether an external creature or a self-representation, which the user attends to as an indicator of their breathing. These designs are typically evaluated in terms of embodiment, measuring whether the temporal coupling between breathing and visual response enhances body ownership, felt agency, and awareness of the respiratory signal (Monti et al. 2020a; Wald et al. 2023).

We contrast these with designs that embed respiratory feedback in the surrounding virtual environment rather than in a discrete object. DEEP (Bossenbroek et al. 2020; van Rooij et al. 2016) and Flowborne (Rockstroh et al. 2021) use diaphragmatic breathing to drive locomotion through immersive landscapes, rewarding paced breathing with movement and environmental change (Bossenbroek et al. 2020; Rockstroh et al. 2021). In a seated variant, Blum et al. (2020) mapped exhalation onto colour changes in flowers, trees, and rocks, keeping the feedback environmental but locally anchored to scene objects. A further group of environmental designs couples not the breathing signal directly but a slower downstream variable, typically heart rate variability, to ambient scene properties. In Blum et al. (2019) and *Heart Garden* (Rook et al. 2026), increasing coherence or heart rate variability gradually transforms

139 the scene. These designs create calming feedback loops, but because the environment
 140 changes through a slower physiological proxy rather than respiration itself, moment-
 141 to-moment respiratory agency is attenuated. The locomotion-based designs provide
 142 immediate agency, but mainly as a progression mechanic rather than as a way of mak-
 143 ing surrounding space itself feel breath-coupled. In our design, we sought to synthesize
 144 these two approaches by mapping the user’s breathing state onto an environment that
 145 expands and contracts with respiration. This provides immediate feedback and a sense
 146 of agency while presenting breathing as an ambient property of the surrounding space
 147 rather than as a distinct object requiring continuous attention. Specifically, the envi-
 148 ronment is composed of particles that respond to the user’s breath such that inhalation
 149 expands the space outward and exhalation contracts it inward. In this way, changes
 150 in internal lung volume are mapped onto changes in external space. This mapping is
 151 closely related to what [Goral et al. \(2024b\)](#) describe as exteroceptive–interoceptive
 152 sensory substitution, in which internal bodily signals are translated into external sen-
 153 sory events that remain experientially linked to the body rather than appearing as
 154 separate readouts. By mirroring the felt expansion and contraction of the thorax,
 155 our system was intended to preserve the natural structure of breathing experience
 156 and thereby function as an intuitive interface that can be understood with minimal
 157 explicit instruction ([Antle et al. 2009](#)). This choice was also informed by evidence
 158 from embodied cognition showing that people judge quantities as larger after inhala-
 159 tion than after exhalation ([Belli et al. 2021](#)), which may indicate a possible, though
 160 still speculative, link between respiratory state and the perception of space. More
 161 broadly, this embodied metaphor appears across a range of immersive biofeedback
 162 systems that map respiration onto expanding and contracting visual stimuli ([Wald
 163 et al. 2023](#); [Goral et al. 2024b](#); [Barton et al. 2024](#); [Stepanova et al. 2020](#)). Two exist-
 164 ing systems are conceptually closest to the approach presented here. [Barton et al.
 165 \(2024\)](#) placed participants inside a virtual orb covered in abstract fractal patterns that
 166 expanded and contracted as participants spread and closed their arms in synchrony
 167 with their breathing; however, the interaction was structured as an explicit movement-
 168 synchronisation task rather than as an environment that responds to breathing on
 169 its own. [Goral et al. \(2024b\)](#) introduced exteroceptive–interoceptive sensory substitu-
 170 tion in a system that translated respiratory signals into dynamic visual and auditory
 171 stimuli intended to feel continuous with the breathing body; however, their imple-
 172 mentation externalized the signal through projection rather than embedding it in an
 173 immersive three-dimensional environment. Our system was designed to combine the
 174 strengths of these two approaches by pairing real-time environmental responsiveness
 175 that affords immediate respiratory agency with feedback distributed across surround-
 176 ing space rather than concentrated in a single object or task. Rather than relying on
 177 semantic cues such as naturalistic scenery, skyboxes, or rendered objects, we pursued
 178 a bottom-up design in which the sense of surrounding space emerges from relations
 179 among many simple particles, drawing on Gestalt principles such as proximity, com-
 180 mon fate, and continuity ([Wagemans et al. 2012](#)) so that their dynamic arrangement,
 181 and specifically their responses to expansion and contraction, conveys depth and close-
 182 ness dynamically within an abstract particle-based environment. This design rationale
 183 also motivates the broader question of whether coupling an internal bodily signal such
 184

as respiration to the surrounding sense of space can also reshape the sense of self, which we consider next through a review of relevant VR experiments.

1.3 VR interventions testing the malleability of the self

Virtual reality has been used extensively to investigate the malleability of bodily self-experience through multisensory manipulation. In full-body illusion paradigms, synchronous visuotactile stimulation between a participant’s physical body and a seen virtual body can shift perceived self-location and induce a sense of ownership over the virtual body (Lenggenhager et al. 2007; Petkova and Ehrsson 2008). Later work showed that interoceptive signals can serve a similar synchronising function. Presenting a virtual body or silhouette that flashes in synchrony with the participant’s heartbeat enhances body ownership and alters self-location even in the absence of tactile stimulation (Aspell et al. 2013; Suzuki et al. 2013). Respiratory signals have been used in a similar way. For example, synchronising a virtual avatar’s chest movements, viewed from a first-person perspective, with the user’s breathing enhances body ownership (Monti et al. 2020a). Similarly, mapping respiratory signals onto the expansion and contraction of a virtual bubble, viewed from a third-person perspective, enhances agency and body awareness (Wald et al. 2023). Collectively, these studies show that a range of sensory modalities, including touch, breathing, and cardiac activity, can be used to reinforce visual feedback and thereby strengthen the sense of embodiment toward a virtual object or avatar from both first-person and third-person perspectives (Lenggenhager et al. 2007; Petkova and Ehrsson 2008; Aspell et al. 2013; Suzuki et al. 2013; Monti et al. 2020a; Wald et al. 2023).

A related but distinct line of research concerns peripersonal space, the region immediately surrounding the body where the brain integrates multisensory signals in body-centered coordinates to maintain a dynamic boundary between self and environment (Serino 2019; Bufacchi and Iannetti 2018). This boundary is not fixed; it extends with tool use (Maravita and Iriki 2004), shifts with emotional valence and social context (Bufacchi and Iannetti 2018; Serino 2019), and interacts with interoceptive aspects of bodily self-representation (Tsakiris et al. 2011a; Azzalini et al. 2019). Recent virtual-reality work further suggests that embodiment and multisensory manipulations can reshape peripersonal space in VR (Petrizzo et al. 2024; Karakoc et al. 2025). While the embodiment paradigms reviewed above manipulate the felt relationship between self and a visual body representation, peripersonal space research highlights that self-experience also depends on how the brain represents the spatial relationship between the body and the surrounding environment (Serino 2019; Bufacchi and Iannetti 2018).

Notably absent from both lines of research are paradigms that intervene through the surrounding environment rather than through a representation of the body itself. In most VR studies of embodiment and self-location, self-related experience is altered by manipulating how the body is represented, with any effects on perceived space arising indirectly through changes in how the body is seen in space (Guy et al. 2023). Although VR research has also developed techniques that manipulate space itself, such as scaling, translation, and redirected movement, these are typically used to support navigation and interaction rather than to investigate self-experience (Abtahi et al. 2022). By contrast, the present approach couples respiration not to a virtual body or

discrete object, but to the spatial extent of the surrounding environment itself. This may engage the boundary between body-centred and environment-centred space in a way that body-focused paradigms do not.

There is a broad consensus that the sense of self is multidimensional (Millière 2020; Taves 2020), and the present study assessed three complementary pictographic measures. These were body-boundary salience (Dambrun 2016), spatial frame of reference (Hanley and Garland 2019), and perceived self-size (Vidal et al. 2025). At the same time, the intervention was expected to affect not only self-experience, but potentially the broader structure of subjective experience. The present study therefore also considered its effects in relation to psychometric frameworks from altered-states research, which informed both measurement and the design rationale of the VR system described next.

1.4 Breathwork as an experiential design space for virtual reality

Embedding a breathing protocol in a responsive virtual environment introduces specific design constraints. Different breathing practices produce different respiratory patterns, including variations in the pace, duration, and force of inhalation and exhalation. Visual feedback coupled to these dynamics must therefore avoid becoming dizzying, claustrophobic, or overwhelming. Visual feedback coupled to either level must avoid becoming disorienting. In VR, large-amplitude radial scene motion, particularly in the visual periphery, is a documented source of cybersickness (Chang et al. 2020; Pöhlmann et al. 2021), and contracting environments can elicit involuntary defensive postural responses even in non-phobic users (Kodithuwakku Arachchige et al. 2022). These risks are compounded when the user is in a heightened or unfamiliar experiential state, given that negative affect and arousal are associated with increased VR sickness severity (Kaufeld et al. 2022). The choice of breathing protocol therefore shapes not only what physiological and experiential effects to expect, but also what the visual environment can safely and comfortably do in response.

High-ventilation breathwork practices such as holotropic breathing, conscious connected breathing, and cyclic hyperventilation involve sustained increases in breathing rate and depth that, over extended sessions, induce pronounced physiological changes, including respiratory alkalosis and reductions in cerebral blood flow, and can give rise to marked changes in subjective experience (Lewis-Healey et al. 2024; Kartar et al. 2025; Fincham et al. 2024). Participants report experiences of oceanic boundlessness, in which the felt boundary between self and world dissolves into a sense of unity or merging with the surroundings; visionary restructuralisation, involving vivid geometric imagery, complex visual scenes, and synesthetic blending of sensory modalities; and dread of ego dissolution, characterised by a felt loss of control over one’s thoughts and sense of identity (Bahi et al. 2024a; Havenith et al. 2025a; Lewis-Healey et al. 2024). These experiential dimensions overlap with those reported under medium- to high-dose psilocybin as measured by the 11D-ASC (Studerus et al. 2010; Havenith et al. 2025a), and are often accompanied by emotional catharsis and strong bodily sensations such as tingling, dizziness, and autonomic arousal (Havenith et al. 2025a; Lewis-Healey et al. 2024; Fincham et al. 2024, 2025). However, the rapid, large-amplitude chest

excursions characteristic of high-ventilation protocols would produce correspondingly rapid expansion and contraction of a breath-coupled particle environment, and the accompanying physiological arousal and altered experiential state would coincide with precisely the visual conditions most likely to provoke cybersickness and spatial discomfort. For these reasons, we chose to facilitate slow-paced breathing rather than high-ventilation breathwork in the present system.

Slow-paced breathing occupies a different region of this subjective and affective space. Its established effect profile is characterised by relaxation, reduced anxiety, increased comfort, and calm alertness, supported by parasympathetic dominance and enhanced heart rate variability (Zaccaro et al. 2018; Balban et al. 2023; Fincham et al. 2023). Slow breathing is also one of the most common attentional anchors in guided meditation practice (Gerritsen and Band 2018; Zaccaro et al. 2018), and evidence from experienced pranayama practitioners indicates that very slow nasal breathing can induce altered states of consciousness characterised by altered body perception, distortions in time sense, unusual salience, increased joy, and reduced anxiety (Zaccaro et al. 2022). The mild but documented non-ordinary experiential profile of slow-paced breathing provided a low baseline against which any augmentation by the visual environment would be easier to detect, while keeping the environment’s spatial dynamics within a comfortable perceptual range.

Augmenting subjective experience is clinically relevant because experiential intensity during altered-state interventions predicts therapeutic outcomes (Roseman et al. 2018; Romeo et al. 2025; Havenith et al. 2025a). We chose a slow-paced protocol precisely because its mild baseline provides sensitivity for detecting whether we can enhance specific attributes of the subjective experience through VR. To characterise this experiential profile and compare it with existing breathwork and psychedelic literatures, we adopted the 11D-ASC (Studerus et al. 2010; Dittrich 1998), the standard instrument across these fields. Its subscales capture the phenomenological features (e.g., boundary dissolution, changed imagery, synesthetic blending) that a breath-coupled environment might selectively amplify, and can be summarised into three higher-order dimensions reflecting positive, distressing, and perceptual effects (Stocker et al. 2025). Because we also examine how these dimensions evolve over the course of a breathing session, we include psychometric time-series ratings, which we refer to as temporal experience tracers (Jachs 2021). We match the tracers one-to-one with the 11D-ASC dimensions to compare global ratings with time-resolved intensity profiles of the same experiential constructs. The motivation to enhance subjective experience is also reflected in the aesthetic design of the visual environment, which we describe next.

1.5 Particle dynamics as a bioresponsive design medium

Viscereality is a bioresponsive VR system developed through an iterative design process, parts of which have been reported previously with respect to its design rationale, technical implementation, and interaction mechanics in human-computer interaction contexts (Fejer et al. 2025; Fejer 2026). The present study provides its first controlled empirical evaluation, testing whether the system’s particle dynamics measurably shape subjective experience during breathwork. As described above, the system maps lung

volume onto a surrounding particle field through interoceptive-exteroceptive sensory substitution (Goral et al. 2024b). Inhalation expands the environment outward, whereas exhalation contracts it inward. Existing implementations of this type of coupling typically use modelled surfaces, either presented as localized targets linked to the user (Wald et al. 2023) or projected onto the walls of a physical room (Goral et al. 2024b). In both cases, the user watches a representation change size, but a uniform surface that grows or shrinks does not, on its own, strongly convey that the surrounding space itself is approaching or receding; that impression depends on motion-based depth cues that arise when many elements move relative to one another (Rogers and Graham 1979). We designed Viscereality as a particle system specifically to afford this. The environment is composed of approximately 2,500 particles distributed on a geodesic icosphere, and the sense of spatial expansion and contraction arises from Gestalt grouping among these elements, specifically proximity, common fate, and continuity (Wagemans et al. 2012), rather than from a texture mapped onto a surface. Because each particle independently carries multiple visual properties (position, size, colour, transparency, motion trajectory), the same elements that produce the breath-coupled spatial dynamics can simultaneously encode additional biofeedback channels, such as heart rate variability coherence (Fejer 2026), without requiring separate visual layers. And because inter-particle interaction rules, specifically oscillator coupling across neighbours, generate emergent aesthetic properties such as dynamic symmetry and geometric coherence, these can be parameterised independently of the breathing mechanics, making the aesthetic texture of the environment an experimentally controllable variable. Throughout the breathing cycle, the particle field remains continuously active. Surface-wave oscillations, chromatic cycles, and shape fluctuations continue during inhalations, exhalations, and both breath-hold phases. The within-VR manipulation therefore does not switch visual motion on or off; rather, during breath holds the high-symmetry condition transiently aligns these ongoing particle cycles into coherent geometric organisations, whereas the asymmetric condition leaves the same dynamics desynchronised. To our knowledge, no existing interoceptive-exteroceptive sensory substitution system takes this particle-based approach. Aesthetics in VR are often treated as incidental to intervention design, yet recent work identifies them as among the strongest predictors of emotional response and experience quality in immersive environments (Marcolin et al. 2021; Cheiran et al. 2025). In Viscereality, aesthetics are not a cosmetic layer but emerge from the same particle dynamics that carry the respiratory coupling, drawing on a longer history of expressive and aesthetic design in particle-based systems applied in visual effects animation.

1.5.1 Particle systems as a generative medium

Particle systems were introduced by Reeves (1983) to represent visual phenomena that have no fixed boundary and no stable shape, including fire, smoke, clouds, and water. In his formulation, each particle is born with stochastically assigned properties (position, velocity, colour, size, transparency, lifetime), moves and changes over time, and eventually vanishes. The overall form is never fully specified; it emerges from the aggregate of many simple elements, each governed by a small set of controllable parameters. This is the property that makes particle systems useful as a bioresponsive

medium because each particle independently carries multiple visual channels, so a single population of elements can simultaneously respond to physiological signals through spatial position, colour shifts, size, transparency, or motion trajectory without requiring separate visual layers for each mapping. Building on this work, Reynolds (1987) introduced interaction between elements, enabling coordinated collective behaviour to emerge from local neighbour rules, whereas in a standard particle system each element responds only to global parameters and physics without awareness of its neighbours. Reynolds specified three local rules. They were to avoid collision with nearby elements, match their velocity, and maintain proximity to them. This became foundational for the ability of 3D animation to produce the fluid, coordinated group motion recognisable in flocking birds, schooling fish, or stampeding herds, all arising bottom-up from neighbour interactions without a centrally scripted choreography.

1.5.2 Particle systems as an affective medium

Research on abstract emotion representation shows that affective meaning can be conveyed through low-level perceptual parameters of abstract expressions (De Rooij et al. 2013). In particular, affective valence can be mapped through semantically minimal features such as motion speed and trajectory smoothness (Bartram and Nakatani 2009; Pollick et al. 2001; Burger and Toiviainen 2020), angular or rounded shape contours (Blazhenkova and Kumar 2018; Aronoff 2006), and line orientation (Damiano et al. 2023). In conceptual terms, Khatin-Zadeh et al. (2019) argue that motion provides an effective vehicle for representing emotion because both unfold dynamically over time, allowing changes in affective state to be expressed through movement, direction, and position in space. This claim has also been borne out empirically in swarm robotics, where the collective motion of distributed abstract disc-shaped agents has been shown to alter the observer’s state of flow and sense of time relative to observing a single moving agent (Kaduk et al. 2024). Dietz et al. (2017) found that higher speed in swarm motion increased perceived emotional intensity, while smooth trajectories, relative to jittery motion, were associated with more positive affect. Extending this approach, Santos and Egerstedt (2021) translated trajectory and motion into swarm choreographies for Ekman’s six basic emotions (Ekman 1993). Rounded and smooth configurations marked positive or low-arousal states, while angular, clustered, or dispersive formations with fast or slow trajectories marked negative and higher-arousal states. Participants identified the intended emotions above chance (Santos and Egerstedt 2021). What makes these findings from swarm robotics relevant here is that their expressive effects arise from relational motion among many simple elements rather than from the form of any individual agent. The same principle holds in *Viscerality*, where local coupling among particles governs trajectory speed, smoothness, synchrony, clustering, and dispersal, producing higher-order collective motion patterns from which affective qualities may emerge. The particle field can therefore be treated as a medium for conveying affective states through abstract visual dynamics rather than semantic or figurative cues.

1.5.3 Particle systems as an aesthetic medium

Beyond serving as a medium for affective mapping, we designed the particle field as an aesthetic means of rendering affective states through abstract spatiotemporal organization. Research on abstract visual art suggests that non-figurative forms evoke aesthetic and affective responses through the isolation of simple features, the grouping of many elements into emergent wholes, and the amplification of perceptual qualities through Gestalt grouping principles (de Rooij et al. 2013; Ramachandran and Hirstein 1999). In our particle system, these same grouping principles also help convey depth when approximately 2,500 ring-shaped particles expand and contract around the user. Because each particle is visually minimal, higher-order forms are not pre-given but emerge through local relations among neighbouring elements, allowing circles, lattices, polygons, and other geometric organisations to arise through changes in three-dimensional location. The appearance of such forms can itself become aesthetically salient, as the resolution of an initially indeterminate field into a coherent Gestalt is known to produce pleasurable moments of insight or ‘aesthetic Aha’ moments (Muth and Carbon 2013; Spehar and van Tonder 2017; Spee et al. 2024). In Viscerality, this made it possible to translate oscillator-level control into changes in overall visual order, symmetry, and coherence without moving to figurative imagery.

Another design influence was the geometric and kaleidoscopic qualities of hallucinatory and psychedelic imagery, which are often marked by geometric patterns, bilateral symmetry, and perceptual ambiguity (Klüver 1966; Billock and Tsou 2012; Lawrence et al. 2022; Rubin et al. 2010). This motivated our use of ring-shaped particles, as their overlaps can give rise to distinctive emergent patterns during the breathing cycle. Because these qualities emerge from shifting relations among many simple particles rather than from fixed symbolic forms, they remain open in meaning and foreground a defining affordance of the medium, namely its capacity to continually reorganise and transform its aesthetic structure over time.

In the context of VR design, Glowacki (2024) terms such non-figurative scenes a weak representational aesthetic, in contrast to photorealistic depictions that render scenes and bodies as recognizable real-world entities. In practice, this aesthetic has been applied in VR scenarios designed to induce altered states of consciousness through the representation of participants’ bodies as luminous, semi-corporeal avatar bodies composed of particle clouds with mergeable body boundaries, contributing to the malleability of self-experience and altered subjective states (Glowacki et al. 2020, 2022; Glowacki 2024; Vidal et al. 2025). Our approach adopts a related design aesthetic while focusing more specifically on diminishing the felt boundary between self and environment.

To evaluate the particle system’s capacity to manipulate aesthetic properties, we operationalized symmetry as an additional independent variable that varied across conditions. Previous studies have linked visual symmetry to more positive affect and perceptual fluency (Bertamini et al. 2013; Pecchinenda et al. 2014; Makin et al. 2012), while synchrony in group movement has been linked to greater enjoyment and more positive affective appraisal (Vicary et al. 2017; Mogan et al. 2017). We therefore asked whether higher degrees of symmetry would bias experience toward more positively valenced states in the context of Viscerality (Kuhn 2024; Johnson 2023). We were also

interested in whether participants would be able to detect and discriminate between the two conditions.

1.6 Present Study

The present study provides the first controlled empirical evaluation of Viscerality with naive participants. We used a preregistered within-subjects design with $N = 39$. Each participant completed three 10-minute audio-guided box-breathing sessions in counterbalanced order. These were a high-symmetry condition, in which positive oscillator coupling transiently aligned ongoing particle oscillations into dynamic geometric coherence during breath-hold phases; an anti-symmetry condition, in which negative coupling produced a visually irregular particle field throughout; and a dark-screen control (see Section 2.3.4). The two particle conditions used identical particles, breath-coupled spatial dynamics, colour palette, and audio guidance, differing only in oscillator coupling parameters. The central within-VR contrast therefore concerned the aesthetic organisation of an otherwise identical breath-responsive scene.

We assessed four substantive outcome domains together with a methodological blinding assessment. Breathing adherence was quantified as the phase-locking value between the observed respiratory signal and an ideal reference trajectory. Self-related experience was assessed with three pictographic measures, namely body-boundary permeability (Dambrun 2016), spatial frame of reference (Hanley and Garland 2019), and perceived self-size (Vidal et al. 2025). Altered states of consciousness were assessed retrospectively with the 11D-ASC (Studerus et al. 2010; Dittrich 1998). Blinding efficacy between the two active VR conditions was assessed retrospectively with a two-stage procedure. Before unmasking, participants completed 12 probe questions covering condition-specific, shared, and sham features. After being told that the two active scenes differed, they indicated in which order the high-symmetry and anti-symmetry scenes had been presented, together with confidence ratings. These responses were used to estimate condition-identification blinding with a standard single-group Bang blinding index and an exploratory confidence-weighted directional variant (Bang et al. 2004).

We hypothesised that, relative to the dark-screen control, the bioresponsive VR conditions would increase breathing adherence and altered-state intensity, and would also influence the three self-related measures, which were included more exploratorily because prior evidence for environment-centred respiratory manipulations remains limited. Within VR, we further expected the high-symmetry condition to bias experience toward more positively valenced dimensions than the anti-symmetry condition.

2 Methods

2.1 Participants

Participants were recruited from the University of Konstanz psychology student participant pool and received course credit for participation. The study used a within-subject design in which each participant completed all three conditions in one of six counterbalanced block-order permutations, with a preregistered target of 6 completers

per permutation and 36 total. We did not systematically assess prior experience with virtual reality, meditation, breathwork, or psychedelics, so the sample should be understood as a general psychology participant-pool sample rather than as a formally screened novice cohort. We tested 47 participants; 8 were excluded based on breathing signal quality (phase-locking value (PLV) outlier criterion described below), yielding a final sample of $N = 39$ (6–7 per permutation), preserving approximately equal representation across all six condition sequences. Among the included participants, mean age was 20.8 years; 37 were female (mean age 20.7 years) and 2 were male (mean age 23.0 years).

516

517 2.2 Design, Conditions, and Procedure

518

Participants were seated in a bean bag, wore noise-blocking headphones, held a controller against their abdomen, and completed all breathing blocks, questionnaires, and outcome measures fully immersed in virtual reality. The three-session procedure was automated end-to-end in the headset: once the headset was put on, participants followed in-VR instructions for the breathing blocks and all post-block outcome measures while remaining seated, with minimal experimenter intervention beyond initial setup and debriefing. Before the experimental sequence, participants completed a 1-minute guided breathing practice block for acclimation, familiarizing them with controller placement and the paced-breathing rhythm. Each participant then completed three conditions, with session order assigned automatically by random selection from the six counterbalanced block-order permutations. Because the dynamic symmetry manipulation was embedded in the software-controlled scene logic and the condition sequence was assigned automatically at runtime, neither the participant nor the experimenter knew which condition would appear in which session. In all conditions, participants followed a 10-minute audio-guided box-breathing exercise (sourced from a publicly available recording) consisting of repeated 4-4-4-4 cycles with four seconds each of inhalation, breath hold, exhalation, and breath hold. In the high-symmetry condition, the bioresponsive particle environment formed dynamic symmetric patterns during breath-hold phases. In the asymmetric condition, visual features of the particles remained asymmetric to one another throughout. In the dark-screen control condition, the particle environment was not displayed; participants viewed a dark field containing only a central fixation marker that also provided breathing-performance feedback (Section 2.3.4). We refer to this condition as the “dark-screen control” throughout. After each block, participants completed the full assessment battery (questionnaires and pictographic scales) while remaining in virtual reality. Figure 1 summarizes the full procedure, including the repeated condition loop and blinding sequence, using the same condition-color mapping as the Results figures.

546

547

548

549

550

551

552

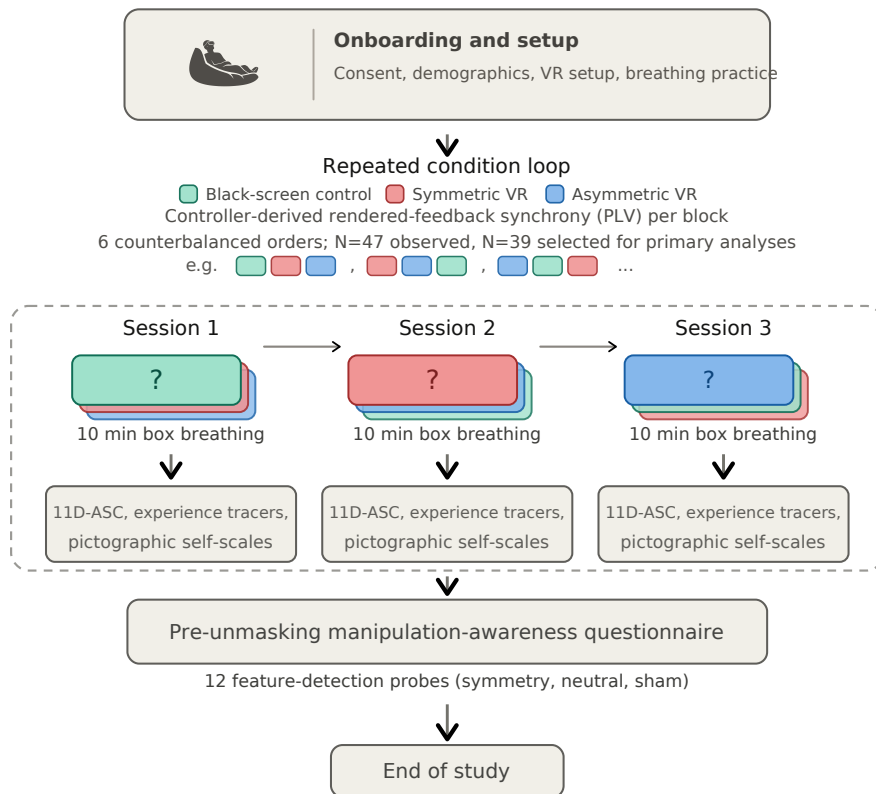


Fig. 1 Study procedure. After onboarding, each participant completed three 10-minute box-breathing sessions (dark-screen control, symmetric VR, asymmetric VR) in one of six counterbalanced orders, with the full assessment battery administered in VR after each block. A blinding questionnaire followed the third block before the study concluded.

2.3 Virtual-Reality Intervention

Viscerality is a bioresponsive virtual-reality system that maps respiratory signals to dynamic visual feedback within an immersive 3D particle environment (Fejer et al. 2025; Fejer 2026). A video summary of the system is available online.¹ The system runs on a Meta Quest 3 headset (standalone, untethered) and was built in Unity, with CPU-optimized parallel processing (Unity Job System with Burst compilation) maintaining a stable 72 Hz frame rate. The two active virtual-reality conditions used the same visual environment, audio guidance, and logging pipeline; they differed only in a single engine parameter controlling oscillator coupling dynamics. The dark-screen control used identical audio guidance and data logging but omitted the particle visualization. We focus here on the experiment-relevant functional logic; fuller technical

¹<https://www.youtube.com/watch?v=Qm-lmC-xYCI>; archived at <https://osf.io/4enf8>

599 and implementation details of the Viscereality engine are described in the prior system
600 papers (Fejer et al. 2025; Fejer 2026).

601

602 2.3.1 Breath Detection

603

604 Breath detection follows the approach introduced by Flowborne (Rockstroh et al.
605 2021), using the VR controller’s positional tracking to infer respiratory displacement
606 without additional sensors. During an initial calibration, the user places the controller
607 against the lower abdomen to establish a tracking axis. Each frame, the algorithm reads
608 the controller’s position and orientation, projects motion onto this breathing-relevant
609 axis, and updates a short-term versus long-term smoothed motion estimate. The dif-
610 ference between these smoothed signals functions as a breathing trend signal. Positive
611 values indicate inhalation, negative values indicate exhalation, and near-zero values
612 indicate a pause. Samples are rejected as tracking artifacts if controller rotation is too
613 abrupt or if the inferred trend is implausibly large. The resulting classification (inhale,
614 exhale, pause, or bad tracking) drives the visual feedback pipeline described below
615 (Figure 2). Detailed implementation choices and parameterization are documented in
616 the Viscereality technical descriptions (Fejer et al. 2025; Fejer 2026).

617

618 2.3.2 Sphere Radius and Visual Feedback

619

620 The user is positioned at the center of a large particle-covered sphere. The primary
621 feedback loop couples sphere radius to breathing. Inhalation causes the sphere to
622 expand outward, exhalation causes it to contract inward, and breath holds main-
623 tain the current radius. Functionally, the controller-derived breathing-state label is
624 converted into a signed expansion/contraction command whose magnitude is scaled
625 to the configured radius range and the intended inhale/exhale timing. That com-
626 mand updates a target sphere radius, and the rendered radius is then smoothed with
627 a damped interpolation step (0.25 s smooth time; overshoot prevention enabled). In
628 practice, this means participants do not see a raw controller-motion trace; they see a
629 bounded, temporally stable visual signal that preserves breathing direction while sup-
630 pressing jitter. Full implementation details of the radius controller are described in
631 the prior Viscereality system reports (Fejer et al. 2025; Fejer 2026).

632

633

634

635

636

637

638

639

640

641

642

643

644

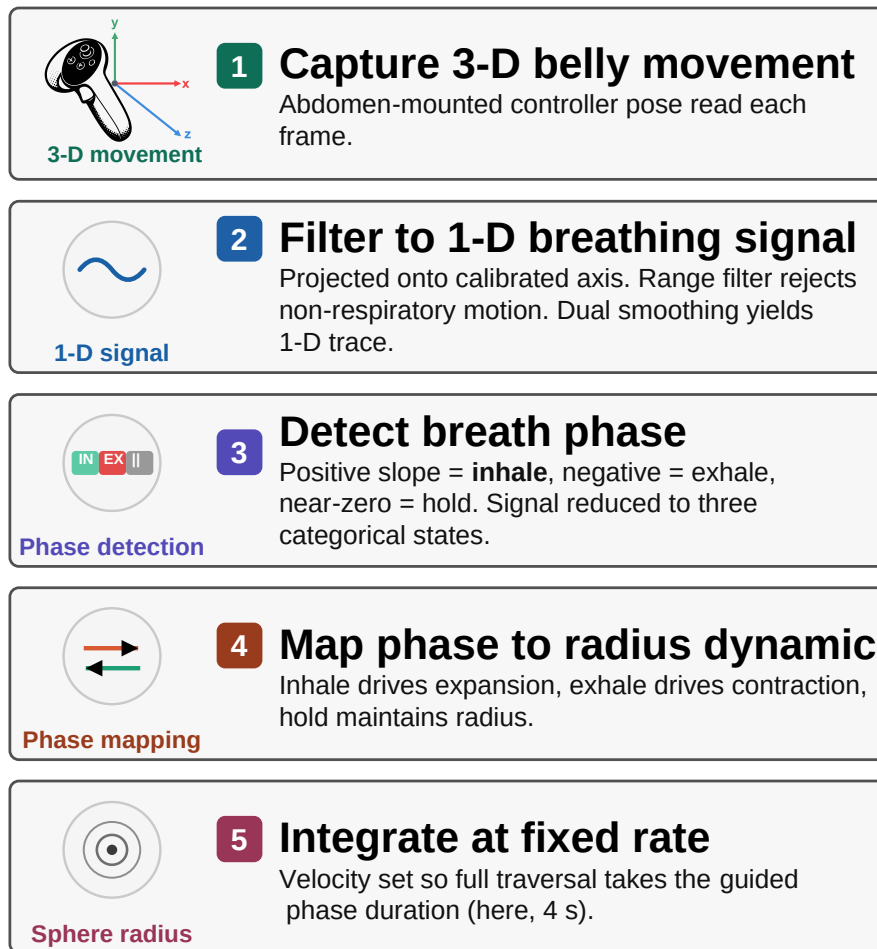


Fig. 2 Signal processing pipeline from abdomen-mounted controller motion to rendered sphere radius. Raw 3D movement is projected onto a calibrated breathing axis, classified into inhale, exhale, or hold phases, and mapped onto expansion and contraction dynamics that update the visible sphere at a fixed velocity.

The sphere surface is composed of approximately 2,500 view-aligned ring particles distributed on an icosphere mesh for uniform geodesic spacing. Each particle is rendered with a radial gradient that produces depth cues through overlap, and during expansion and contraction, particles generate cone-like tracer effects that reinforce the perception of three-dimensional movement. Surface wave oscillations, structured noise, and unsynchronized chromatic cycles maintain continuous visual dynamics across all breathing phases, preventing the environment from appearing static during breath holds.

2.3.3 Oscillator Network and Condition Manipulation

The second layer of the system implements a Kuramoto oscillator network across the sphere’s vertices (Acebrón et al. 2005; Rodrigues et al. 2016). Each particle has an internal oscillation phase (analogous to a clock angle) and interacts with its neighbors through coupling weights that determine whether nearby particles tend toward synchronization or away from it. If the coupling weight is positive, neighbors are pulled toward shared timing (phase attraction). If negative, the interaction becomes *anti-coupling*, and neighbors are pushed away from alignment (phase repulsion).

The network uses a two-tier neighbor structure reflecting the icosphere topology. Most vertices have 6 immediate (tier-1) neighbors and 12 next-nearest (tier-2) neighbors, while 12 vertices at the original icosahedron poles have 5 tier-1 neighbors. At each update step, each oscillator advances according to its own intrinsic rhythm plus a neighborhood interaction term. In plain terms, the algorithm compares an oscillator’s current phase with those of its neighbors and then adjusts it toward or away from them depending on the sign and magnitude of the tier-specific coupling weights. This is the standard Kuramoto logic (phase-offset-dependent attraction/repulsion), implemented here with two neighbor tiers and a global coupling gain, and updated in parallel each frame together with particle position, size, and color. The formal implementation and parameterization are described in the prior Viscereality publications (Fejer et al. 2025; Fejer 2026).

The two active conditions differed in how coupling weights were assigned. In the high-symmetry condition, tier-1 and tier-2 coupling weights were interpolated dynamically as a function of a coherence scalar (0–1), together with a coherence-dependent frequency scaling term. Verbally, low coherence pushes the system toward anti-coupled, irregular dynamics, whereas higher coherence progressively shifts interactions toward phase attraction and more synchronized motion. As a result, particle motion, color, and size periodically coordinate into coherent geometric patterns, such as flower-of-life-like lattices, during breath-hold phases before dissolving back into disorder when breathing resumes.

In the asymmetric condition, the system did not use this dynamic interpolation. Instead, coupling weights were held at fixed negative values (anti-coupling), so local interactions were permanently biased toward phase repulsion and the particle field remained visually irregular throughout. The manipulation was thus dynamical rather than geometric. The same scene, particles, and audio were used in both conditions, and only the oscillator interaction rules changed.

2.3.4 Breathing Performance Indicator

All three conditions displayed a minimal breathing performance indicator at the center of the participant’s field of view. This consisted of a white pentagon ring whose individual line segments scaled with sphere radius: at half-volume (midpoint between full inhale and full exhale), the pentagon was fully visible; as the participant inhaled or exhaled toward the extremes, the segments progressively shortened and disappeared entirely at maximum and minimum volume. The indicator thus functioned as an unobtrusive loading-bar-style monitor of breathing excursion. Participants were instructed

that the goal was to make the pentagon disappear completely on each inhalation and exhalation, confirming sufficient breathing depth.

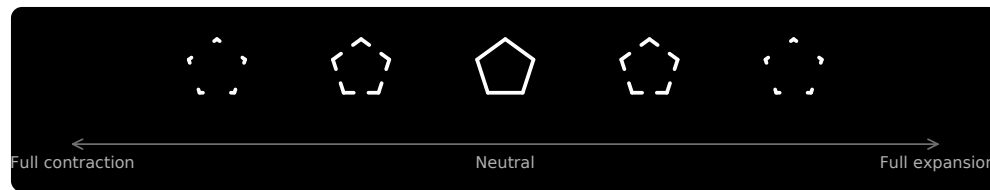


Fig. 3 Central breathing-performance indicator shown in all three conditions. The pentagon ring was fully visible at the neutral sphere radius and progressively disappeared toward full contraction or full expansion, providing condition-matched feedback about breathing excursion.

2.4 Outcome Measures

We retained the five preregistered measurement blocks used in the assessment battery, but organize them here into four recurring outcome families that also structure the Results section. These were adherence to box breathing, self-related measures, altered states of consciousness, and the blinding questionnaire. Within the altered-states family, we present the derived 3D-ASC_r composites first as a dimensionality-reduced overview of the same 11D-ASC data, followed by the underlying 11D-ASC questionnaire.

2.4.1 Adherence to Box Breathing

Adherence quantifies how closely each participant’s breathing followed the instructed 4-4-4-4 box-breathing rhythm across each 10-minute block. The preregistered plan was to compute this from discrete phase labels produced by the breath-detection algorithm (Section 2.3.1), which classifies each frame as inhale, exhale, hold, or bad tracking. The intended metric was phase-duration error, defined as the absolute deviation from the 4-second target per breathing phase, averaged across valid cycles per condition, with participant exclusion if fewer than 70% of cycles were valid or if mean error exceeded 2 standard deviations above the sample mean.

The recorded phase sequences did not yield the expected clean succession of 4-second phases. Rather than discrete blocks of inhale, hold, exhale, and hold, the recorded data consisted of highly fragmented segments interspersed with brief transitional or ambiguous intervals. Computing phase-duration error from this data required first reconstructing plausible box-breathing cycles from the raw phase stream, a step that was not anticipated at preregistration. We explored several reconstruction strategies, including rule-based merging of adjacent micro-segments and temporal smoothing algorithms, but each approach introduced its own arbitrary parameter choices, such as minimum segment duration thresholds, smoothing window widths, and rules for combining adjacent same-phase fragments, that could materially influence the resulting error estimates. Because these post-hoc binning decisions carried more researcher degrees of freedom than the original preregistered metric implied, we adopted an alternative approach.

We quantified adherence as the phase-locking value (PLV) between two continuous time series. One was the sphere-radius signal that participants actually saw as visual feedback during the breathing session. The other was an ideal reference radius trajectory derived from the audio-instruction timestamps for perfect 4-4-4-4 breathing. PLV measures how consistently the phase relationship between two signals remains stable over time. A value near 1 indicates that the participant’s breathing closely tracked the instructed rhythm throughout the block. A value near 0 indicates little consistent alignment. This approach circumvents the phase-reconstruction problem entirely, because it evaluates alignment at the level of the continuous visual feedback signal rather than attempting to recover discrete phase boundaries from a fragmented controller trace. PLV thus captures the same underlying construct as the preregistered metric, namely how closely breathing tracked the instructed rhythm, while avoiding the measurement artifacts that any post-hoc phase-segmentation strategy would have introduced. Figure 4 illustrates the relationship between observed and reference trajectories for a representative participant; the fragmented phase-label structure visible in the lower bars of each panel motivates the PLV-based approach described above.

Because PLV does not decompose into per-cycle validity counts, the preregistered 70%-valid-cycle plus 2-SD mean-error exclusion rule could not be transferred directly. Outlier exclusion instead used a lower-tail global IQR rule on condition-level PLV values, elevated to participant-level exclusion if any condition was flagged. Supplementary Section S2 describes PLV computation and quality screening in detail.

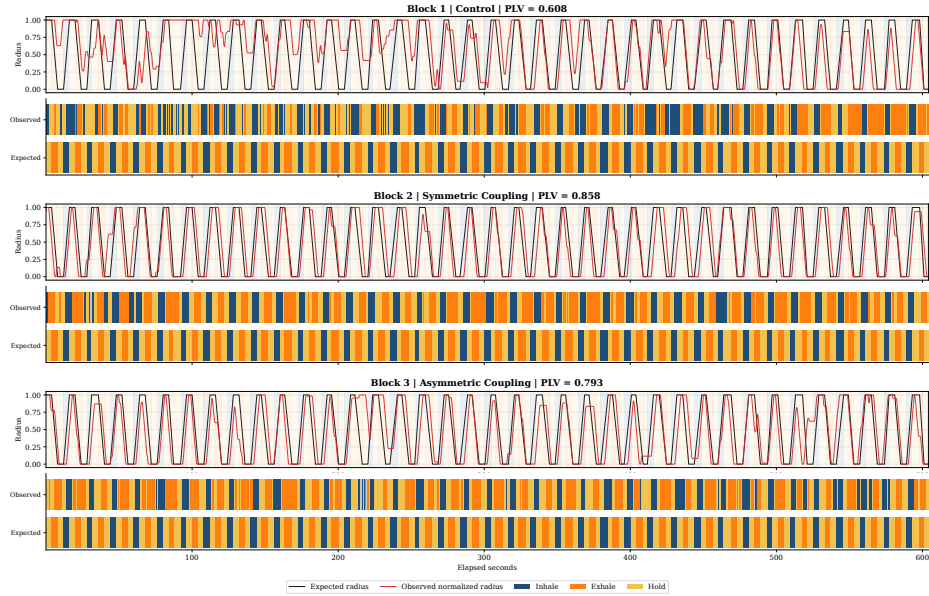


Fig. 4 Breathing dynamics for a representative participant (P01) across all three conditions. Observed sphere-radius traces (orange) are overlaid on the ideal 4-4-4-4 reference (blue), with per-block PLV values shown at right. Fragmented phase-label bars below each time series illustrate the segmentation noise that motivated the switch from phase-duration error to PLV as the adherence metric. Equivalent figures for all participants are provided in the OSF repository.

2.4.2 Self-Related Measures

Three single-item pictographic instruments assessed self-related experience after each condition. The Perceived Body Boundaries Scale (Dambrun 2016) presents body silhouettes ranging from sharply delineated to fully diffuse; participants select the image best matching their felt boundary permeability. The Spatial Frame of Reference Continuum (SFoRC; Hanley et al. 2020) depicts a figure in concentric spatial zones from body-centered to environment-encompassing, indexing the degree to which the spatial sense of self extended beyond the body. The Small Self Scale (Vidal et al. 2024) presents figures decreasing in size relative to their surroundings, capturing awe-like self-diminishment. All three yield a single ordinal score per condition. Each pictograph was recreated as a scalable vector graphic for VR presentation, using the published figures as references and refining the SVG source with large language models to preserve the graded structure of the source scale. The OSF reproducibility package (<https://osf.io/64mwk/>) provides the recreated stimulus set, administered item keys, and study materials; Supplementary Section S5 provides the corresponding reproducibility map.

2.4.3 Altered States of Consciousness

The preregistered temporal experience tracers provide psychometric time-series ratings of the same experiential dimensions. Because their results do not add explanatory value beyond the questionnaire results in this study, we treat their analyses as exploratory and report their complete methods, results, and figures in Supplementary Section S4.

3D-ASCr composites. The revised 3D-ASCr higher-order factor model became available after preregistration. It provides a parsimonious contemporary summary of the same preregistered 11D-ASC responses by organizing them into Positive, Distressing, and Perceptual Effects. This level of aggregation matches our broad theoretical interest in positively versus negatively valenced experience, while retaining a perceptual dimension, and avoids overinterpreting lower-order dimensions for which no specific predictions were made. We therefore use the 3D-ASCr composites as the main-text summary. Supplementary Section S3 reports complete analyses of all 11 preregistered 11D-ASC dimensions.

11D-ASC Questionnaire. We assessed altered states with the 11-Dimensional Altered States of Consciousness Rating Scale (11D-ASC; 42 items, 0–100 visual analogue scales), a validated lower-order factorization of the OAV/APZ framework (Studerus et al. 2010; Dittrich 1998; Hovmand et al. 2024; Prugger et al. 2022). The 11 subscales were Experience of Unity (EU), Spiritual Experience (SE), Blissful State (BS), Insightfulness (IS), Disembodiment (DIS), Impaired Control and Cognition (ICC), Anxiety (ANX), Complex Imagery (CI), Elementary Imagery (EI), Audio-Visual Synesthesia (AVS), and Changed Meaning of Percepts (CMP). Participants completed one condition-specific rating after each block.

875 2.4.4 Blinding Questionnaire

876 Blinding was assessed in two phases. In the pre-unmasking phase, participants were
877 told that the study investigated how subtle differences in visual aesthetics might influ-
878 ence perception in virtual reality. We constructed a 12-item ad hoc masked-comparison
879 questionnaire for this study. For each item, participants chose whether the feature was
880 stronger in Scenario 1, stronger in Scenario 2, equal in both scenarios, or absent in both
881 scenarios. The chance baseline was 0.25. Scenario labels referred to the two virtual-
882 reality conditions in encounter order. The item set covered three hidden domains.
883 These were manipulation-relevant symmetry/coordination cues that genuinely differed
884 between the two VR conditions, matched comparison features that were present sim-
885 ilarly in both conditions and therefore should be judged as equal, and foil (sham)
886 features that were not present in either condition and therefore should be judged as
887 absent. There were four items per domain.

888 In shorthand, the four symmetry-related items asked whether one scene more
889 strongly showed transient coordination and ordering of the particle field, including
890 particles moving as a coordinated group, becoming more organized or structured at
891 certain moments, showing wave-like coordinated motion-and-color changes, or shift-
892 ing from chaotic to ordered movement during breath holds. The four neutral matched
893 items asked about features that were present in both VR conditions to a similar degree,
894 namely particles becoming brighter during breath holds, leaving faint motion trails
895 during exhalation, remaining in gentle ongoing motion across the surface, or appearing
896 to rise and fall like ocean waves. The four sham items were constructed as absent cross-
897 modal or pulsing foils, asking about the music becoming quieter, the music seeming to
898 slow down, particle motion and color changing in response to the music, or particles
899 showing a size-based pulsing effect. To keep administration uniform while avoiding an
900 obviously grouped structure, items from all three domains were intermixed in a sin-
901 gle randomized order that was generated once in advance and then presented in that
902 same order to all participants.

903 In the post-unmasking phase, participants were told that one condition featured
904 brief visual synchronization during breath holds while the other remained irregular
905 throughout. They indicated which session was more symmetric via a three-option
906 forced choice (*first*, *second*, or *I don't know*), followed by a confidence rating.

908 2.5 Preregistration and Reproducibility

909 The study was preregistered on AsPredicted (#262,545). The unchanged registry
910 record, all deidentified participant-level data, statistical analysis scripts and out-
911 puts, figure-generation scripts, and study materials were uploaded to OSF as a
912 reproducibility package (<https://doi.org/10.17605/OSF.IO/64MWK>). We retained all
913 preregistered design elements and outcome families. The preregistered analysis plan
914 specified a repeated-measures ANOVA with condition as the within-subjects factor and
915 presentation order as a covariate for each outcome, two planned contrasts (pooled VR
916 versus dark-screen control, and symmetric versus asymmetric), and parallel Bayesian
917 analyses. Adherence was to be quantified as phase-duration error with a cycle-validity
918 exclusion rule, as described and motivated in Section 2.4.1. We deviated from this
919
920

plan in several respects. Supplementary Section S1 summarizes the deviations from preregistration and their rationale.

Adherence metric. Phase-duration error was replaced by PLV because the recorded phase-label data were too fragmented to yield reliable per-phase error estimates without introducing arbitrary binning decisions (see Section 2.4.1 for a full account). The associated exclusion rule was changed from 70%-valid-cycle plus 2-SD mean-error to a lower-tail global IQR rule on condition-level PLV values, because PLV does not decompose into per-cycle validity counts.

Omnibus condition test. The preregistered RM-ANOVA with condition and order was replaced by an order-adjusted mixed-effects likelihood-ratio test comparing a model with condition and block-position order against an order-only model, with participant as random intercept. The realized counterbalanced design was more naturally represented by condition-specific block-position covariates than by a single omnibus order term, and this formulation preserves the intended within-subject and order-adjusted comparison without forcing an approximation to the realized order structure. A condition-only repeated-measures ANOVA without the order covariate is retained in the OSF outputs as a sensitivity check.

Planned contrasts. The two preregistered planned contrasts were implemented as standalone paired t -tests, which yield the same paired contrast statistics as ANOVA follow-ups in this within-subject setting. These serve as the confirmatory anchor for all multi-test outcome families in the Results section, and for significant planned contrasts in the altered-state families we additionally report paired common-language effect sizes (CLES), defined as the proportion of participants whose score was higher in the VR condition than in the dark-screen control (McGraw and Wong 1992).

Robustness and multiplicity. Because Shapiro–Wilk checks indicated non-normality for several condition-level and difference-score distributions, Wilcoxon signed-rank tests and Friedman omnibus tests were added as distribution-free robustness checks. The preregistration did not specify a multiple-comparison correction strategy. We applied Benjamini–Hochberg false-discovery-rate (FDR) correction within each multi-test outcome family (11 ASC dimensions, 3 derived 3D-ASCr dimensions, 3 self-related variables). Main-text confirmatory interpretations are anchored in the planned contrasts with uncorrected p values. FDR-adjusted q values and the full robustness-check output are reported in the OSF repository and are mentioned in the main text only when they materially qualify the confirmatory pattern.

3D-ASCr composites. The revised 3D-ASCr scoring scheme (Stocker et al. 2025) became available after preregistration. Because these composites are deterministic averages of the preregistered 11D-ASC subscale scores, they do not introduce additional analytic degrees of freedom and are presented as a dimensionality-reduced summary rather than as separate confirmatory endpoints (see Section 2.4.3).

Bayesian analyses. Bayesian analyses were preregistered and run in parallel on the same included sample ($N = 39$). Omnibus Bayes factors compared a point-null model without condition (H_0 : outcome $\sim$ order + participant random intercept) against a model that additionally included condition (H_1 : outcome $\sim$ condition + order +

participant random intercept), using the BayesFactor R package with explicit prior settings (fixed-effect $r = 0.50$, continuous-covariate $r = \sqrt{2}/4 \approx 0.354$, random-effect nuisance $r = 1.00$; Morey and Rouder 2023; van Doorn et al. 2021; Kruschke 2021). For the two planned paired contrasts, we compared H_0 ($\delta = 0$) against two-sided H_1 ($\delta \neq 0$) using a Bayesian paired t -test with the default Cauchy prior on standardized effect size ($r = \sqrt{2}/2 \approx 0.707$; Morey and Rouder 2023; van Doorn et al. 2021). Pre-registered one-sided ordered checks (symmetric > asymmetric > dark-screen control) were restricted to EU, BS, and AVS. We report BF_{10} values (and $\text{BF}_{01} = 1/\text{BF}_{10}$ when the data favor the null) as relative evidence rather than as posterior probabilities or effect sizes (van Doorn et al. 2021; Kruschke 2021). Posterior median standardized effect sizes with 95% credible intervals and prior-robustness checks are reported in the OSF repository.

Blinding analyses. Pre-unmasking accuracy was scored using the documented answer key for the four symmetry, four neutral, and four sham items and across all 12 items. Mean accuracy was compared with the 0.25 chance baseline using Wilcoxon signed-rank tests, with Benjamini–Hochberg FDR correction across the four comparisons. Post-unmasking guesses were classified as correct (R), incorrect (W), or uncertain (U) from the participant’s selected active-VR session and condition order. The standard single-group Bang blinding index was calculated as $\text{BBI} = (R - W)/(R + W + U)$ (Bang et al. 2004; Kolahi et al. 2009), with a participant-bootstrap 95% confidence interval. Effective blinding was considered established only if the entire interval fell within $[-0.2, 0.2]$ (Qin et al. 2024; Kolahi et al. 2009). We also report an exploratory confidence-weighted directional index, calculated as the confidence-weighted proportion of correct directional guesses minus 0.5. The standard Bang interpretation bands were not applied to this modified index. To test whether pre-unmasking symmetry accuracy predicted subsequent condition identification, we calculated a point-biserial correlation between the 0–4 symmetry score and final-guess correctness, classified as correct versus incorrect or uncertain, with an exact two-sided permutation p value.

Exploratory analyses. We fit condition $\times$ PLV interaction models to test whether breathing synchronization moderated subjective effects.

3 Results

Across the four recurring outcome families, condition effects were consistently driven by the contrast between virtual-reality conditions and the dark-screen control. No outcome showed a reliable symmetric-versus-asymmetric difference. We report the results in the same family order used in Methods: adherence, self-related measures, altered states of consciousness, and the blinding questionnaire. For all multi-test families, confirmatory interpretations are anchored in the preregistered planned contrasts evaluated as paired t -tests, with the order-adjusted mixed-effects omnibus noted for each family as context. Wilcoxon signed-rank robustness checks, Friedman omnibus tests, and FDR-adjusted q values are documented in the OSF repository and are mentioned here only when they materially qualify the paired t pattern.

3.1 Sample and Exclusions

The dataset contained records from all 47 tested participants. Condition-level PLV values from all participants were used to apply the quality-screening procedure described in Section 2.4.1 and illustrated in Figure 5. This procedure excluded 8 participants and yielded a final analytic sample of $N = 39$. The 39 included participants each contributed a complete within-subject triad and constituted the final sample used for all inferential analyses.

3.2 Outcome Families

3.2.1 Adherence (PLV)

To test whether visual biofeedback improved adherence to the guided box-breathing rhythm, we compared PLV across conditions. Mean PLV was $M = 0.787$ ($SD = 0.141$) for the dark-screen control, $M = 0.836$ ($SD = 0.127$) for symmetric VR, and $M = 0.860$ ($SD = 0.094$) for asymmetric VR. The planned pooled-VR-versus-dark-screen-control contrast confirmed this difference, $t(38) = 3.47$, $p = .001$, $d_z = 0.56$, 95% CI for the mean PLV difference $[0.026, 0.097]$. The symmetric-versus-asymmetric contrast was not significant, $t(38) = -1.13$, $p = .267$, $d_z = -0.18$. The order-adjusted mixed-effects omnibus supported a condition effect, $\chi^2(2) = 12.66$, $p = .002$. The corresponding Bayesian planned contrasts showed that the pooled-VR-versus-dark-screen-control difference was about 24.1 times more likely than no difference ($BF_{10} = 24.11$), whereas the symmetric-versus-asymmetric data were about 3.2 times more likely under a no-difference model ($BF_{01} = 3.22$).

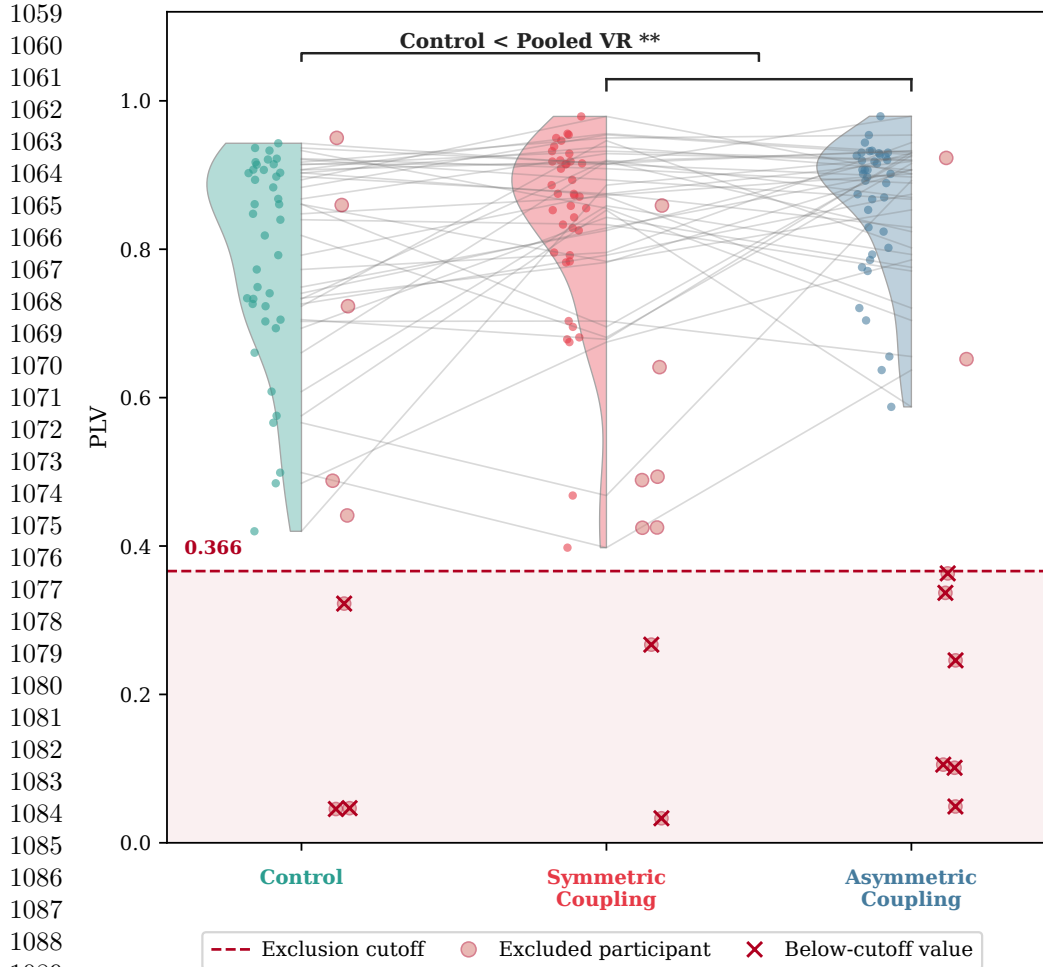


Fig. 5 Controller-derived feedback synchrony (PLV) by condition for the selected sample ($N = 39$), with condition-wise distributions and paired individual trajectories in gray. The dashed red line marks the pooled lower-tail IQR exclusion cutoff (0.366); hollow circles and red crosses indicate flagged participants and their below-cutoff values, respectively. If any single condition fell below the cutoff, the participant was excluded entirely from all analyses across all conditions.

3.2.2 Self-Related Measures

To assess whether the breath-responsive virtual environment shifted subjective self-representation, we compared three pictographic instruments across conditions. Only the spatial frame-of-reference continuum showed a condition effect. Participants reported a more spatially extended sense of self during virtual-reality conditions, with $M = 3.15$ ($SD = 1.41$) for the dark-screen control, $M = 4.18$ ($SD = 1.82$) for symmetric VR, and $M = 4.26$ ($SD = 1.73$) for asymmetric VR. The planned pooled-VR-versus-dark-screen-control contrast confirmed the effect, $t(38) = 4.28$, $p < .001$, $d_z = 0.69$, 95% CI [0.56, 1.57]. The symmetric-versus-asymmetric contrast was not

significant, $t(38) = -0.32$, $p = .752$. The order-adjusted mixed-effects omnibus converged with this pattern, $\chi^2(2) = 20.04$, $p < .001$. Neither perceived body-boundary permeability ($t(38) = -0.17$, $p = .864$) nor small self ($t(38) = 1.39$, $p = .173$) showed a significant VR-versus-dark-screen-control contrast, and no self-related measure showed a significant symmetric-versus-asymmetric contrast.

Bayesian comparisons agreed with this pattern at the planned-contrast level. For spatial frame of reference, the pooled-VR-versus-dark-screen-control contrast was about 204.4 times more likely than no difference ($BF_{10} = 204.44$). Perceived body boundaries instead favored the null for that same contrast ($BF_{01} = 5.71$), while small self remained only weakly informative in the null direction ($BF_{01} = 2.39$), consistent with the absence of a reliable self-related effect outside the spatial-frame measure.

3.2.3 Altered States of Consciousness

We report the altered-state family using the 3D-ASCr composites as a dimensionality-reduced overview and the underlying 11D-ASC subscales. The clearest pattern is stronger positive and perceptual VR-related effects, with a weaker distress-related component and no reliable symmetric-versus-asymmetric difference.

3D-ASCr composites.

Summarizing the 11D-ASC data through the 3D-ASCr composites (see Section 2.4.3 for derivation), the three higher-order dimensions showed the following pattern. Perceptual Effects had $M = 11.41$ ($SD = 12.61$) in the dark-screen control, $M = 27.25$ ($SD = 19.92$) in symmetric VR, and $M = 25.59$ ($SD = 17.24$) in asymmetric VR. Positive Effects had $M = 18.88$ ($SD = 15.39$) in the dark-screen control, $M = 25.64$ ($SD = 19.48$) in symmetric VR, and $M = 26.46$ ($SD = 17.84$) in asymmetric VR. Distressing Effects had $M = 12.33$ ($SD = 10.26$) in the dark-screen control, $M = 17.33$ ($SD = 12.66$) in symmetric VR, and $M = 17.02$ ($SD = 13.18$) in asymmetric VR. All three composites showed significant pooled-VR-versus-dark-screen-control planned contrasts: Perceptual Effects $t(38) = 4.90$, $p < .001$, $d_z = 0.79$; Positive Effects $t(38) = 3.22$, $p = .003$, $d_z = 0.51$; Distressing Effects $t(38) = 2.56$, $p = .014$, $d_z = 0.41$. Expressed on the native 0–100 scale, pooled VR increased Perceptual Effects by 15.0 points, Positive Effects by 7.2 points, and Distressing Effects by 4.8 points. These corresponded to a large effect for Perceptual Effects and moderate effects for Positive and Distressing Effects, with 76.9%, 69.2%, and 76.9% of participants, respectively, scoring higher in VR than in the dark-screen control.

No composite showed a symmetric-versus-asymmetric difference (all $p \geq .357$). Order-adjusted mixed-effects omnibus tests converged for all three composites (Perceptual Effects $\chi^2(2) = 32.83$, Positive Effects $\chi^2(2) = 14.32$, Distressing Effects $\chi^2(2) = 9.11$; all $p \leq .011$). Bayesian omnibus comparisons favored condition models for all three composites, with the strongest evidence for Perceptual Effects ($BF_{10} = 121,810$), followed by Positive Effects ($BF_{10} = 30.49$) and Distressing Effects ($BF_{10} = 3.45$; Figure 6).

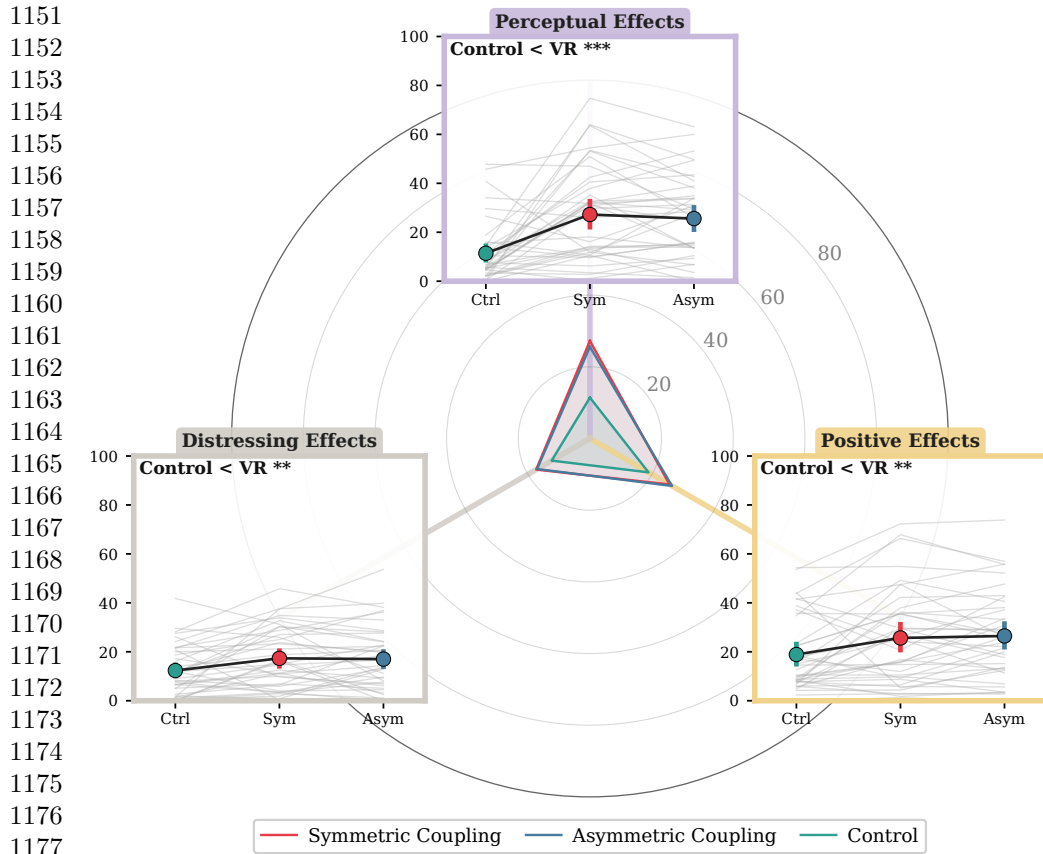


Fig. 6 Condition-wise distributions for the three 3D-ASCr composites (N = 39). All three dimensions increased in VR relative to the dark-screen control, with the largest effect for Perceptual Effects; no symmetric-versus-asymmetric differences emerged.

11D-ASC questionnaire.

Six subscales showed significant pooled-VR-versus-dark-screen-control planned contrasts: Experience of Unity ($t(38) = 3.50, p = .001, d_z = 0.56$), Spiritual Experience ($t(38) = 2.58, p = .014, d_z = 0.41$), Disembodiment ($t(38) = 2.59, p = .014, d_z = 0.41$), Elementary Imagery ($t(38) = 4.71, p < .001, d_z = 0.75$), Audio-Visual Synesthesia ($t(38) = 5.15, p < .001, d_z = 0.82$), and Changed Meaning of Percepts ($t(38) = 3.30, p = .002, d_z = 0.53$). On the native ASC scale, these pooled-VR increases ranged from 6.6 points for Spiritual Experience to 25.6 points for Audio-Visual Synesthesia. Standardized effects ranged from moderate ($d_z = 0.41$ for Spiritual Experience and Disembodiment) to large ($d_z = 0.82$ for Audio-Visual Synesthesia), with 67.9% to 87.2% of participants scoring higher in VR than in the dark-screen control across these six dimensions. Impaired Control and Cognition was significant at the uncorrected level ($t(38) = 2.15, p = .038, d_z = 0.34$) but borderline after within-family FDR adjustment ($q = .060$). Anxiety did not reach significance ($t(38) = 1.92,$

$p = .063$). No subscale showed a significant symmetric-versus-asymmetric contrast (all $p \geq .166$). Condition-wise descriptives for all 11 subscales are reported in the OSF repository.

Order-adjusted mixed-effects omnibus tests converged with the planned-contrast pattern for most subscales, with one divergence. Impaired Control and Cognition showed a marginal mixed-model omnibus ($p = .040$, $q = .062$) but was supported by a non-parametric Friedman robustness check ($p < .001$); this divergence is documented in the OSF repository.

Bayesian planned contrasts supported the same six dimensions. The strongest evidence concentrated in Audio-Visual Synesthesia ($BF_{10} = 2,408$) and Elementary Imagery ($BF_{10} = 670$), followed by Experience of Unity ($BF_{10} = 64.0$), Changed Meaning of Percepts ($BF_{10} = 12.6$), Spiritual Experience ($BF_{10} = 5.60$), and Disembodiment ($BF_{10} = 4.94$). Blissful State, Insightfulness, and Complex Imagery favored the null (BF_{01} range 3.3–8.8), while Impaired Control and Cognition and Anxiety remained inconclusive. Symmetric-versus-asymmetric Bayesian contrasts uniformly favored no difference (BF_{01} range 2.3–5.7; per-dimension values in the OSF repository).

The preregistered one-sided ordering checks were more selective. For the predicted asymmetric-greater-than-dark-screen-control step, the data supported the directional effect for Experience of Unity ($BF_{10} = 87.7$) and Audio-Visual Synesthesia ($BF_{10} = 745.0$) but not for Blissful State ($BF_{01} = 2.68$). For the predicted symmetric-greater-than-asymmetric step, the data favored the null for Experience of Unity and Blissful State ($BF_{01} = 8.07$ and 12.43) and remained inconclusive for Audio-Visual Synesthesia ($BF_{10} = 0.79$).

For descriptive cross-study context, we compared the pooled Viscerality VR profile with four published breathwork datasets that reported compatible ASC summaries: conscious connected breathing (Bahi et al. 2024a), facilitated circular breathwork (Havenith et al. 2025b), high-ventilation breathwork accompanied by music in an MRI/laboratory study (Kartar et al. 2025), and Holotropic Breathwork (Uthaug et al. 2021). Viscerality was lower than Bahi et al. across all three 3D-ASCr composites; it was close to Havenith et al. on Positive and Perceptual Effects but lower on Distressing Effects. Relative to Kartar et al., it showed lower Positive Effects but higher Distressing and Perceptual Effects, and it was higher than Uthaug et al. across all three composites in the extracted summary. These overlays compare raw profile means rather than matched effect sizes and confound breathing technique, session duration, participant experience, music and facilitation, and control structure. They are therefore descriptive only (Figure 8).

3.2.4 Blinding Questionnaire

Pre-unmasking responses showed selective but incomplete sensitivity to the symmetry manipulation. Mean symmetry-item accuracy was 44.2%, equivalent to approximately 1.77 of four items, and exceeded the 25% chance baseline (FDR-adjusted $q < .001$). Neutral-item accuracy was 27.6% ($q = .610$), and sham-item accuracy was 32.7% ($q = .100$); neither was clearly above chance. Across all 12 items, mean accuracy was

1243 34.8% ($q < .001$). Participants therefore detected some genuine symmetry features,
1244 but performance remained well below perfect identification.

1245 Post-unmasking guesses were nearly balanced: 17 participants guessed correctly
1246 (43.6%), 19 guessed incorrectly (48.7%), and 3 were uncertain (7.7%). The standard
1247 Bang blinding index was close to zero ($\text{BBI} = -.051$, 95% bootstrap CI $[-.359, .256]$).
1248 Confidence weighting changed little: the exploratory confidence-weighted index was
1249 $-.016$ (95% bootstrap CI $[-.203, .168]$, $n = 35$ directional responses). Both intervals
1250 crossed zero, indicating no reliable tendency toward either correct or opposite identifi-
1251 cation. However, the standard BBI interval extended beyond the predefined $[-0.2, 0.2]$
1252 blinding region. The results were therefore consistent with effective blinding but too
1253 uncertain to rule out meaningful identification in either direction.

1254 To test whether pre-unmasking sensitivity predicted subsequent condition iden-
1255 tification, we related the 0–4 symmetry score to final-guess correctness, classified as
1256 correct versus incorrect or uncertain. Pre-unmasking symmetry-detection accuracy
1257 was descriptively higher among participants who subsequently identified the correct
1258 condition (2.00 versus 1.59 of four items), but the association was small and incon-
1259 clusive (point-biserial $r = .18$, exact two-sided permutation $p = .33$; Figure 7d).
1260 Thus, better masked detection did not reliably predict the final condition guess in this
1261 sample.

1262
1263
1264
1265
1266
1267
1268
1269
1270
1271
1272
1273
1274
1275
1276
1277
1278
1279
1280
1281
1282
1283
1284
1285
1286
1287
1288

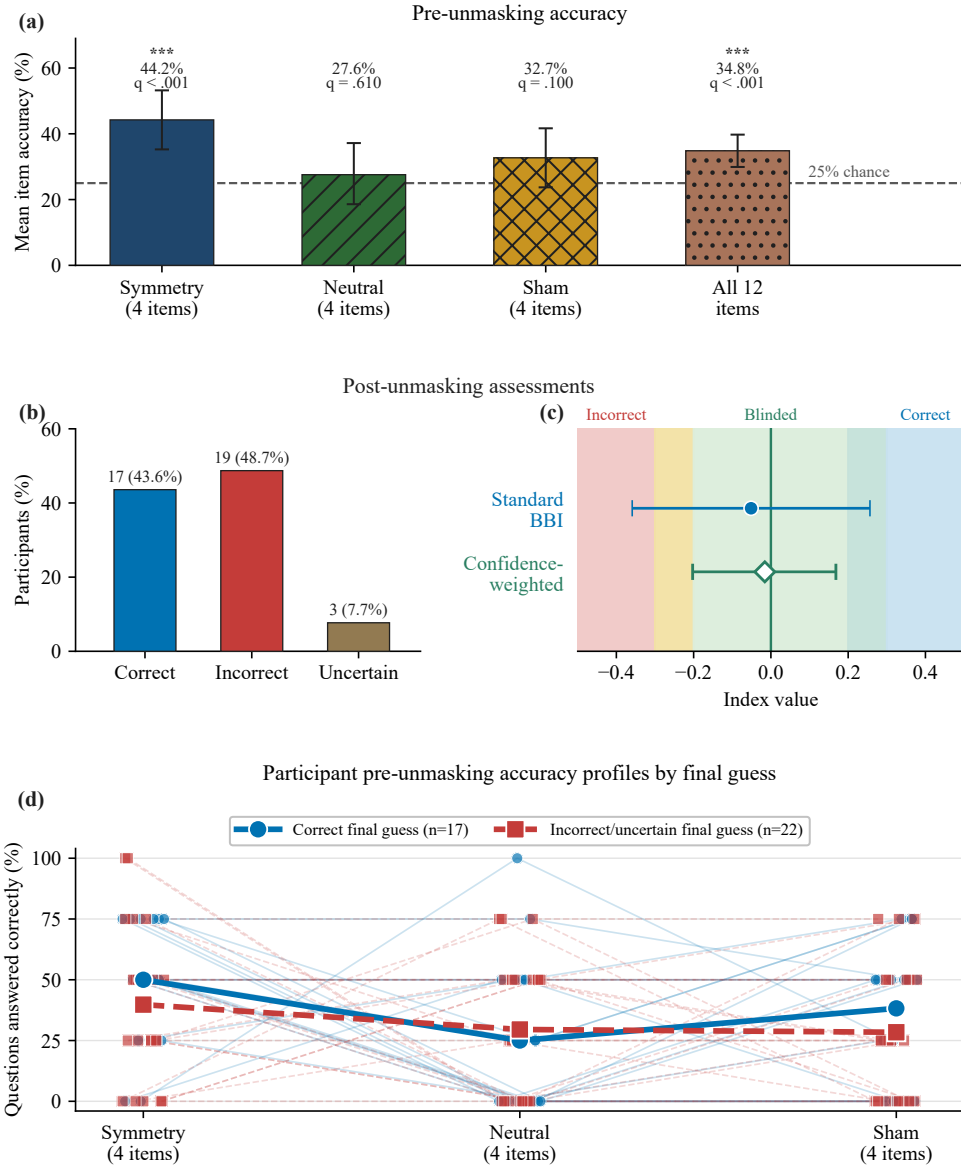


Fig. 7 Blinding and manipulation-awareness summary ($N = 39$). (a) Mean pre-unmasking accuracy for the four symmetry, four neutral, and four sham items and across all 12 items. The dashed line marks the 25% chance baseline; asterisks denote FDR-adjusted $q < .001$. (b) Post-unmasking guesses: 17 correct, 19 incorrect, and 3 uncertain. (c) Standard Bang blinding index (filled blue circle) and exploratory confidence-weighted directional index (open green diamond), with participant-bootstrap 95% confidence intervals. The shaded regions indicate incorrect, blinded, and correct identification and apply only to the standard Bang index. (d) Participant-level pre-unmasking accuracy across the three four-item categories, grouped by final-guess correctness. Thin lines and small markers represent participants; heavier lines and larger markers represent group means. Incorrect and uncertain guesses are combined.

1335 3.3 Exploratory Checks

1336 Exploratory mixed-model checks for order and PLV moderation are reported in the
1337 OSF repository. No FDR-corrected order main effects, condition $\times$ order, condition $\times$
1338 PLV, or condition $\times$ order $\times$ PLV interactions emerged for any ASC, 3D-ASCr, or self-
1339 related outcome family. For PLV itself, there was a main effect of order, $\chi^2(1) = 8.28$,
1340 $p = .004$, but no condition $\times$ order interaction, $\chi^2(2) = 0.37$, $p = .833$, indicating
1341 that block order did not change the between-condition adherence pattern. A planned
1342 linear contrast across session position was significant, $t(38) = -3.16$, $p = .003$, 95% CI
1343 $[-0.100, -0.022]$, indicating that PLV decreased rather than increased across sessions
1344 and therefore aligning more with waning engagement or fatigue than with a practice-
1345 improves account.

1347 4 Discussion

1349 4.1 Summary of findings

1351 This study provided a first controlled empirical evaluation of Viscereality with naive
1352 participants. Relative to audio-guided box breathing in the dark-screen control, the
1353 two VR conditions improved breathing adherence, shifted spatial self-experience
1354 toward a more expanded frame of reference, and produced an altered-state profile
1355 in which perceptual and positive effects outweighed mild distressing effects, with
1356 no adverse events across 47 tested participants. However, the preregistered aesthetic
1357 manipulation that varied the symmetry of the particle dynamics did not reliably dif-
1358 ferentiate the two active VR conditions. Taken together, these findings indicate that
1359 the paradigm produced a structured experiential profile during a brief breathing ses-
1360 sion with naive participants, but stronger control comparisons and objective measures
1361 are needed to determine the specificity and depth of these effects.

1363 4.2 Breathing adherence

1364 Participants in both active VR conditions followed the instructed breathing pattern
1365 more closely than in the dark-screen control. The display responded to participants'
1366 ongoing breathing rather than to the audio guide, so it made respiration continuously
1367 visible without telling participants how well they were matching the instructed pac-
1368 ing. One plausible interpretation is that the breath-responsive environment helped
1369 adherence by keeping participants engaged and by giving them the sense that their
1370 breathing was producing the visual experience. Previous findings are consistent with
1371 this. Breath-coupled VR feedback has been linked to improved focus on breathing and
1372 greater diaphragmatic movement (Blum et al. 2020; Rockstroh et al. 2021), greater
1373 focused attention and less distraction than non-VR biofeedback (Rockstroh et al.
1374 2019), and increased interoceptive sensibility (Goral et al. 2024a). A simpler explana-
1375 tion also remains possible. Guided breathing in the dark-screen control is presumably
1376 less engaging than the same task in an immersive environment, and the adherence met-
1377 ric itself was derived from how closely breathing matched the trajectory expressed by
1378 the visual display. The study would have benefited from a more stringent control, such
1379 as a visually matched scene that does not respond to breathing or one that replays
1380

a prerecorded animation, because the engaging visual environment on its own may have contributed to the effect. This caveat matters because controlled comparisons in the VR breathing literature have not consistently shown VR-based interventions to outperform matched non-VR approaches on clinical outcomes (Cortez-Vázquez et al. 2024; Pancini et al. 2025). The present adherence finding should therefore be treated as preliminary until tested against a more stringent control.

4.3 Self-related experience

Of the three self-related measures, only the spatial frame-of-reference continuum showed a condition effect. We had reason to expect body-boundary salience to shift as well, since this measure decreases reliably during meditation (Dambrun 2016) and, in one study that administered both scales, mindfulness training reduced perceived body boundaries and expanded allocentric spatial framing in tandem, with the boundary change mediating the spatial shift (Hanley et al. 2020). That pattern did not emerge here, perhaps because the two constructs respond to partly different triggers. Body-boundary dissolution appears to shift most reliably during practices that direct sustained attention toward the body itself, such as body-scan meditation (Dambrun 2016), mindfulness training (Hanley et al. 2020), and floatation-REST (Hruby et al. 2024). Spatial frame of reference, by contrast, has been linked not only to body-directed contemplative practice (Hanley et al. 2020) but also to perceived expansion of the physical frame of reference beyond the body (Dambrun 2023). Because Viscerality acts on the surrounding visual-spatial environment rather than directing attention toward the body, it may have engaged the spatial-frame construct through this environmental route without producing the body-focused conditions under which boundary dissolution typically occurs. Of the three scales, spatial frame of reference is also the one most closely aligned with the broader question motivating this research programme, namely whether breath-coupled changes in surrounding space can alter the felt extension of self into the environment. Previous work has shown that mapping respiration onto an externalized virtual avatar can modulate body ownership (Monti et al. 2020b) and self-location (Adler et al. 2014; Betka et al. 2020) through afferent respiratory pathways (Betka et al. 2020). However, this result requires cautious interpretation. The concentric depiction of the SFoRC scale may have resembled the expanding and contracting particle sphere shown in VR, which raises the possibility that participants selected more spatially expansive depictions because the scale imagery matched what they had just seen rather than because their felt self-boundary had shifted. This demand-characteristics concern cannot be ruled out with pictographic self-report alone and represents a clear limitation of the present measurement approach. The finding should accordingly be treated as preliminary pending corroboration from measures that do not share visual features with the stimulus, such as peripersonal space tasks. If confirmed by such measures, the spatial-frame shift would support continued investigation of whether mapping respiration onto surrounding space can modulate the felt extension of self into the environment.

1427 4.4 Altered states of consciousness

1428 Within the context of a 10-minute box-breathing exercise with ambient background
1429 music, *Viscerality* increased all three higher-order dimensions of the Altered States
1430 of Consciousness Rating Scale relative to the dark-screen control, which provided
1431 only a central fixation marker and minimal visual breathing-performance feedback.
1432 Perceptual Effects showed the largest increase, Positive Effects a moderate increase,
1433 and Distressing Effects a smaller increase. None differed reliably between the two
1434 VR mappings that manipulated visual symmetry, and participants showed only lim-
1435 ited feature-level sensitivity to this manipulation without reliable condition-level
1436 identification.

1437 The fine-grained subscale decomposition and item-content analysis reveal a more
1438 specific experiential profile (Supplementary Section S3.4). The perceptual shift pri-
1439 marily concerned simple patterns, colors, lights, and flashes, together with impressions
1440 that sound changed visual shapes or colors. These elementary visual phenomena closely
1441 matched features supplied directly by the particle environment, so their increase partly
1442 reflects overlap between the intervention and the questionnaire criteria. Participants
1443 also attributed visual changes to sound even though the same background music was
1444 presented in every condition and did not control the display. One plausible interpre-
1445 tation is that the richer and more dynamic visual stream provided more perceptual
1446 material that could be bound to, or retrospectively attributed to, concurrent sound,
1447 rather than producing audio-visual synaesthesia in the stricter sense. A similar dis-
1448 tinction has been raised in psychedelic research, where LSD increased spontaneous
1449 synaesthesia-like reports that lacked the consistency and inducer specificity associated
1450 with genuine synaesthesia (Terhune et al. 2016). The Positive Effects increase was
1451 concentrated in items reflecting a loosening of separation between self, surroundings,
1452 and time, heightened significance and emotional salience of ordinary objects, sensa-
1453 tions of floating or diminished bodily presence, and moments of awe. The smaller
1454 Distressing Effects increase had the character of diffuse destabilization, encompass-
1455 ing reduced agency, difficulty organizing thought, isolation, unexplained apprehension,
1456 and an altered or threatening quality of the surroundings. Neither constituent distress-
1457 related subscale remained reliable after correction across the fine-grained analyses. We
1458 therefore interpret the composite increase as an accumulation of mild and compati-
1459 ble unpleasant experiences rather than as a pronounced or clearly localized adverse
1460 response.

1461 To interpret the subjective effects beyond their measurement caveats, it helps to sit-
1462 uate them relative to similar induction contexts. The 11D-ASC was designed to enable
1463 comparisons across different techniques and substances (Dittrich 1998; Studerus et al.
1464 2010), and a growing body of cross-study data now supports such contextualization
1465 (Prugger et al. 2022). As shown in Figure 8, we compared the present profile with four
1466 breathwork datasets: conscious connected breathing (Bahi et al. 2024b), facilitated cir-
1467 cular breathwork (Havenith et al. 2025b), high-ventilation breathwork with music in
1468 an MRI/laboratory setting (Kartar et al. 2025), and Holotropic Breathwork (Uthaug
1469 et al. 2021). The marked variation among the profiles displayed in the figure indicates
1470 that subjective-effect ratings depend substantially on the breathing technique and
1471 the wider context in which it is practised. Differences in session duration, participant
1472

experience, music, facilitation, setting, and study design preclude direct comparison, so the figure provides qualitative context rather than evidence of equivalence. More meaningful comparisons require understanding how breathing technique and context shape individual 11D-ASC subscales, and which subscales are most context-dependent. Future studies should combine subscale-level analyses with relevant neurophysiological biomarkers to distinguish shared physiological mechanisms from intervention-specific and contextual effects across breathwork with and without VR.

The comparison in Figure 8 nevertheless highlights an important practical distinction. The reference studies predominantly involved high-ventilation or circular-breathing techniques conducted over longer periods, often with music and facilitator support. Such protocols may be unsuitable for some individuals vulnerable to panic, syncope, hypocapnia, or cardiovascular complications (Fincham et al. 2023), while their duration and facilitation requirements may limit integration into time-constrained clinical settings. By contrast, the present intervention lasted only 10 minutes and produced pronounced Perceptual Effects that fell within the range of some longer breathwork studies, while Distressing Effects remained comparatively low and no adverse events were observed in the present sample. These findings do not establish clinical safety or efficacy, but they identify brief bioresponsive VR paired with slow-paced breathing as a candidate for further study in settings where relaxation-oriented breathing exercises are already established, including neurorehabilitation. Research in clinical populations should determine whether the visual feedback adds meaningful benefit to conventional breathing practice and should directly assess feasibility, accessibility, tolerability, and safety.

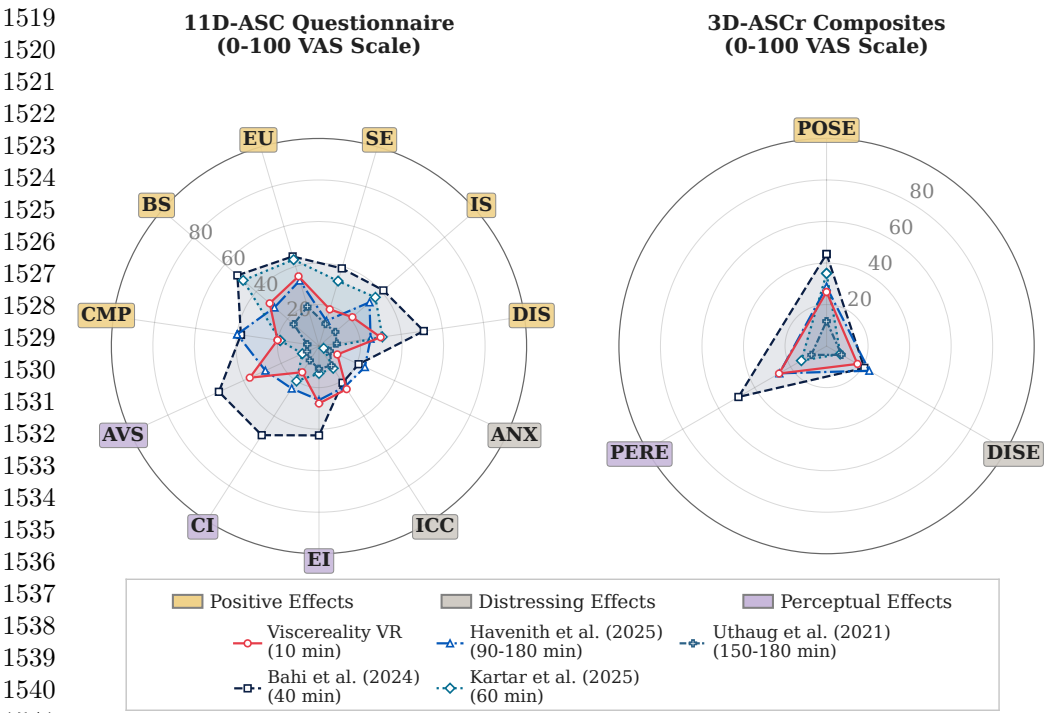


Fig. 8 Descriptive cross-study comparison of altered-state profiles from five named datasets. The left panel shows 11D-ASC subscale means and the right panel shows 3D-ASCr composite means. The Viscerality profile pools the symmetric and asymmetric VR conditions. Lines show raw 0–100 profile means, not standardized effect sizes; differences in technique, duration, sample, music, facilitation, and study design preclude inferential comparison. Legend order is purely graphical and does not imply a numbered study sequence.

Looking closer at the differences across subscales, our VR intervention produced somewhat higher scores on Experience of Unity, Spiritual Experience, Blissful State, and Disembodiment, while circular breathwork produced roughly twice the intensity on Insightfulness and Changed Meaning of Percepts. Bayesian analyses confirmed that these latter two dimensions were largely unaffected by the addition of VR relative to the dark-screen control. One qualitative difference that may partially account for this pattern is where visual awareness is directed. Traditional breathwork typically involves eye shades (Havenith et al. 2025b; Grof 2019) and frames the experience as an inward-directed process, with attention oriented toward bodily sensation (Grof 2019; Rhinewine and Williams 2007). Our VR paradigm directs visual awareness outward, toward a surrounding environment that responds to the breath. The somewhat higher unity and disembodiment scores in our VR intervention may in part reflect this difference, though the many other ways the two paradigms differ make it difficult to say with certainty. Insightfulness and Changed Meaning of Percepts, where circular breathwork produced roughly twice the intensity, capture reflective and integrative processes. In psychedelic research, more challenging experiences have been found to

be associated with greater retrospectively attributed meaningfulness and personal significance (Barrett et al. 2016), and it is plausible that a similar dynamic operates in breathwork, where working through a prolonged and at times distressing session gives rise to a sense of having gained something from the experience. Such effects would be unrealistic to expect after 10 minutes of relaxation-oriented breathing, nor was this the design intention. The present system was designed as a brief, pleasant, self-contained experience for augmenting breath, and in our view we achieved that.

4.5 Blinding and the null symmetry result

The blinding results distinguish masked feature-level sensitivity from post-unmasking condition identification. Before unmasking, symmetry-item accuracy exceeded chance, while neutral and sham accuracy did not. Participants therefore detected some genuine symmetry features, but only imperfectly. After unmasking, correct and incorrect guesses were nearly balanced, and both blinding indices were close to zero. Correct guessers had descriptively higher symmetry accuracy than incorrect or uncertain guessers, but the association was small and inconclusive ($r = .18$, exact $p = .33$). Overall, the results were consistent with effective blinding, but the wide standard BBI interval prevented a conclusive determination.

The fact that participants partially detected the symmetry manipulation at the feature level yet showed no net correct identification at the condition level may also point toward a potential explanation for why no outcome family showed a meaningful difference between the symmetry and anti-symmetry conditions. The geometric coherence unique to the symmetry condition emerged only during breath-hold phases, which nominally occupy half the breathing cycle. But the primary task demand was to drive the expansion and contraction of the particle sphere through inhaling and exhaling, and participants' attention was likely oriented toward this respiratory coupling rather than toward the aesthetic character of the holds. The breath holds remained visually active, as the particle field continued its turbulent oscillations throughout, but these were not the phases where participants felt they were performing the task. The effective attentional exposure to the symmetry difference was therefore considerably less than the nominal 50% of session time. The blinding data compound this ambiguity. The near-zero point estimate could reflect successful blinding, but its wide interval precludes a definitive conclusion; it could also reflect that the manipulation was too subtle or too brief to register as an experientially meaningful difference. The same perceptual subtlety that enables within-modality blinding may also prevent the aesthetic variation from producing measurable shifts in subjective experience. The present design does not provide precise control over the timing, duration, or attentional allocation of exposure to the aesthetic manipulation, and the degree to which task demands distracted from the hold-phase aesthetics is an uncontrolled variable.

Several explanations for the null symmetry result should be weighed together. The manipulation may have been too subtle to produce experiential differences at the level measured by the present outcome scales. The effective exposure may have been too brief, given that the symmetry difference was confined to breath-hold phases and that attention was likely oriented toward the respiratory coupling during inhale and exhale phases. The particular features that were varied, specifically the coupling

1611 strength governing spatial coherence, may not be the features that matter most for
1612 subjective experience in this context. Or the aesthetic difference may genuinely not
1613 produce measurable experiential shifts at this exposure level regardless of attention.
1614 The present data do not distinguish between these accounts, and doing so would
1615 require designs that isolate exposure duration, attentional allocation, and aesthetic
1616 feature type as independent variables.

1617 There are converging indications that the types of features varied in the present
1618 system do contribute to affective processing below the threshold of explicit aware-
1619 ness, which suggests the null result reflects the conditions of exposure rather than
1620 the irrelevance of the features themselves. Visual symmetry carries implicit positive
1621 hedonic valence that dissociates from explicit preference ratings (Ruta et al. 2021),
1622 and the sustained posterior negativity generated by visual regularity scales with the
1623 degree of symmetry regardless of whether participants attend to it (Makin et al. 2023).
1624 Spatial features of surrounding environments, including contour geometry and per-
1625 ceived enclosure, activate reward-related and valence-processing circuitry during both
1626 active aesthetic judgment and passive viewing (Vartanian et al. 2013, 2024; Vessel and
1627 Rubin 2021). Interoceptive signals, including respiratory and cardiac rhythms, modu-
1628 late body representation and self-related experience through largely implicit processes
1629 (Tsakiris et al. 2011b), and spatial perception engages kinesthetic and proprioceptive
1630 dimensions that extend beyond explicit evaluation (Kwon and Iedema 2022). The ques-
1631 tion for future work is therefore not whether these features matter in principle, but
1632 under what conditions of exposure, attention, and duration they produce measurable
1633 shifts in subjective experience within immersive environments.

1634

1635 4.6 Limitations

1636

1637 To probe the efficacy of our VR system as a novel altered-state induction method,
1638 a psychometric measurement scale less reliant on visual descriptors would have been
1639 more conducive to evaluating this capacity, although similar critiques exist around
1640 other instruments such as the Mystical Experience Questionnaire, which has been
1641 criticized for reliance on a religious framing (Sanders and Zijlmans 2021). To confirm
1642 whether the questionnaires and pictographic scales capture real differences reflec-
1643 tive of the constructs they are aimed to measure, namely the system’s capacity to
1644 induce altered states of consciousness and extend subjective self-boundaries, further
1645 evidence through objective markers is needed, preferably ones less dependent on the
1646 visual modality to remain orthogonal to the manipulation. Furthermore, evaluating
1647 the system’s aesthetic parameters requires more reliable experimental controls, for
1648 instance over exposure time and attentional allocation to these features, and more fine-
1649 grained differentiation of what the manipulations are intended to assess. The study
1650 was conducted on Western first-year psychology students, hence the utility of our VR
1651 paradigm, for instance within clinical contexts, is limited pending further testing on
1652 tolerability in the wider population.

1653

1654

1655

1656

4.7 Future directions

The Viscereality system was designed with the aim of inducing experiences that enhance the plasticity of self-representation and dissolve the felt boundary between self and environment, which we situate within the broader phenomenology of altered states of consciousness. The spatial frame of reference finding lends preliminary support for the system’s capacity to shift this boundary, but as with any retrospective self-report measure, it requires corroboration from objective measures that do not depend on the participant’s post-hoc interpretation of their experience. Our future work will use peripersonal space tasks as the primary objective outcome measure. Peripersonal space indexes the spatial boundary surrounding the self through multisensory integration processes that differ in speed depending on whether stimuli are perceived as near or far from the body (Serino 2019; Bufacchi and Iannetti 2018; Karakoc et al. 2025). The utility of this task is that it uses visuotactile synchrony detection probes which remain independent of the visual intervention, and can be administered during the 4-second breath-hold windows after inhaling and exhaling, capturing shifts in self-boundaries in real time while participants remain immersed. The aim would be to test whether the system’s core mapping, in which inhalation expands surrounding space and exhalation contracts it, produces corresponding expansion and contraction of the peripersonal space boundary. Reversing this mapping could test whether congruence between respiratory phase and spatial direction is what drives the effect. Demonstrating that respiratory signals can measurably reshape self-boundaries would establish breathing as a window to a new design space in VR, one in which visceral afferents serve as a medium through which experience is shaped, and how vr experiences can be further utilized to enhance the plasticity of self.

The particle system in the present experiment was used to convey the expansion and contraction of surrounding space in response to breathing, but particles can also encode visceral signals directly onto the body or its surroundings, or construct abstract visual configurations that convey affective states through movement, colour, shape, or synchrony. Our manipulation of dynamic particle symmetry through oscillator coupling explored only one such parameter and did not yield observable effects. Our lesson from this is that mapping aesthetic features of particle displays onto affective dimensions requires a more fine-grained approach than a single global variable assessed through altered-state questionnaires. Short aesthetic judgment tasks offer a tractable starting point, following swarm robotics research where trajectory smoothness, speed, synchronization, and spatial coherence map onto separable affective dimensions (Dietz et al. 2017; Santos and Egerstedt 2021). The immersive 3D particle environment affords considerably more degrees of freedom than planar swarm displays, with particles moving through three spatial dimensions under both local neighbour-coupling and global synchronization parameters, and testing specific feature combinations in controlled affective assessments would provide a more solid basis for these mappings than retrospective questionnaires after prolonged sessions. Because the medium is abstract rather than representational, the same particle field could in principle simultaneously encode multiple biosignals and serve as the basis for biofeedback displays that are intuitive to the user and conducive to the affective states the feedback is meant to support.

The present study demonstrates that the particle system was conducive to pleasant experiences during box breathing, with positive and perceptual effects comparable to prolonged facilitator-supported breathwork (Havenith et al. 2025b) at lower distress and no adverse events. Because the system is self-administered and requires only 10 minutes, it could support breathing practice in settings where access to a facilitator is limited or time is scarce. High-ventilation techniques are contraindicated for individuals vulnerable to panic, syncope, or hypocapnia (Fincham et al. 2023), and a slow-paced breath-coupled VR system offers a lower-intensity entry point into breathing as a therapeutic modality. Stroke survivors are one population where this could be relevant, given that respiratory dysfunction and sleep-disordered breathing are common after stroke and associated with poorer recovery (Patrizz et al. 2023; Seiler et al. 2019), while structured respiratory rehabilitation remains underemphasized in clinical practice (Menezes et al. 2016). Slow-paced breathing exercises are already used to retrain respiratory function after stroke (Abdullahi et al. 2024; Yoon et al. 2022), and coupling breathing to an intuitive visual mapping of respiratory depth could engage an additional learning pathway that supports recovery of motor control over respiration. These remain illustrative possibilities. The present sample consisted of a homogeneous group of university psychology students, and testing usability, tolerability, and efficacy in clinical populations is a necessary next step.

1722

1723 4.8 Conclusion

1724

This study demonstrates that Viscereality, a breath-responsive VR environment that maps lung volume onto a surrounding particle field, is conducive to positive experiential states and supports guided breathing practice during brief self-administered sessions with naive participants. In 10 minutes of box breathing, the system shifted spatial self-experience toward a more expanded frame of reference and produced an altered-state profile in which positive and perceptual effects were comparable in magnitude to those reported after prolonged facilitator-supported circular breathwork, while producing less distress and no adverse events across 47 participants.

1733

1734 Statements and Declarations

1735

Funding: The study reported here was funded by the BIAL Foundation through Grant 308/22, “Altered States of Viscereality: Augmenting Breathwork with Bio-Responsive Virtual Reality,” awarded under the 2022 Grants Programme for Scientific Research. Continued development of Viscereality received support from the Baden-Württemberg Stiftung through the Forschungswerkstatt – Deine Idee für Morgen programme for the project “Atemräume: Integration atem- und herzratenbasierter Interaktionen in therapeutische VR-Umgebungen.”

Competing interests: The authors declare no competing interests.

Ethics approval: The Ethics Committee of the University of Konstanz approved the study, which was conducted in accordance with the Declaration of Helsinki.

Consent to participate: All participants provided informed consent before participation.

1748

Consent for publication: Participants consented to the publication of anonymized, non-identifiable data.

Availability of data and materials: The deidentified participant-level data, original preregistration record, statistical analysis outputs, figure-generation scripts, study materials, and reproducibility documentation are available as a reproducibility package at OSF: <https://doi.org/10.17605/OSF.IO/64MWK>.

Code availability: The statistical analysis and figure-generation scripts, exact environment specifications, and replay instructions are available at OSF: <https://doi.org/10.17605/OSF.IO/64MWK>.

Authors' contributions: George Fejer: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. Till Holzapfel: Conceptualization, Methodology, Resources, Software, Validation, Visualization. Taru Hirvonen: Methodology, Resources, Software, Visualization. Anestis Lalidis Mateo: Methodology, Resources, Software. Johannes Blum: Methodology, Resources, Software, Validation. Michael Gaebler: Conceptualization, Project administration, Supervision, Writing – review & editing. Bigna Lenggenhager: Conceptualization, Funding acquisition, Supervision, Writing – review & editing.

References

- Abdullahi A, Wong TWL, Ng SSM (2024) Efficacy of diaphragmatic breathing exercise on respiratory, cognitive, and motor function outcomes in patients with stroke: a systematic review and meta-analysis. *Frontiers in Neurology* 14:1233408. <https://doi.org/10.3389/fneur.2023.1233408>, URL <https://doi.org/10.3389/fneur.2023.1233408>
- Abtahi P, Hough SQ, Landay JA, et al (2022) Beyond being real: A sensorimotor control perspective on interactions in virtual reality. In: *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. ACM, pp 1–17, <https://doi.org/10.1145/3491102.3517706>, URL <https://doi.org/10.1145/3491102.3517706>
- Acebrón JA, Bonilla LL, Pérez Vicente CJ, et al (2005) The kuramoto model: A simple paradigm for synchronization phenomena. *Reviews of modern physics* 77(1):137–185
- Adler D, Herbelin B, Similowski T, et al (2014) Breathing and sense of self: Visuo-respiratory conflicts alter body self-consciousness. *Respiratory Physiology & Neurobiology* 203:68–74. <https://doi.org/10.1016/j.resp.2014.08.003>, URL <https://doi.org/10.1016/j.resp.2014.08.003>
- Ahmed A, Devi RG, Priya AJ (2021) Effect of box breathing technique on lung function test. *Journal of Pharmaceutical Research International* 33(58A):25–31. <https://doi.org/10.9734/JPRI/2021/v33i58A34085>, URL <https://doi.org/10.9734/JPRI/2021/v33i58A34085>
- Alhammad SA (2024) Advocating for action: Exploring the potential of virtual reality in breathing exercise – a review of the clinical applications. *Patient Preference and*

1795 Adherence 18:695–707. <https://doi.org/10.2147/ppa.s451609>, URL [https://doi.org/](https://doi.org/10.2147/ppa.s451609)
 1796 [10.2147/ppa.s451609](https://doi.org/10.2147/ppa.s451609)
 1797
 1798 Antle AN, Corness G, Droumeva M (2009) What the body knows: Exploring the
 1799 benefits of embodied metaphors in hybrid physical digital environments. *Interacting*
 1800 *with Computers* 21(1–2):66–75. <https://doi.org/10.1016/j.intcom.2008.10.005>, URL
 1801 <https://doi.org/10.1016/j.intcom.2008.10.005>
 1802
 1803 Aronoff J (2006) How we recognize angry and happy emotion in people,
 1804 places, and things. *Cross-Cultural Research* 40(1):83–105. [https://doi.org/10.1177/](https://doi.org/10.1177/1069397105282597)
 1805 [1069397105282597](https://doi.org/10.1177/1069397105282597), URL <https://doi.org/10.1177/1069397105282597>
 1806
 1807 Ashhad S, Kam K, Del Negro CA, et al (2022) Breathing rhythm and pattern and
 1808 their influence on emotion. *Annual Review of Neuroscience* 45:223–247. [https://doi.](https://doi.org/10.1146/annurev-neuro-090121-014424)
 1809 [org/10.1146/annurev-neuro-090121-014424](https://doi.org/10.1146/annurev-neuro-090121-014424)
 1810
 1811 Aspell JE, Heydrich L, Marillier G, et al (2013) Turning body and self inside out. *Psy-*
 1812 *chological Science* 24(12):2445–2453. <https://doi.org/10.1177/0956797613498395>,
 1813 URL <https://doi.org/10.1177/0956797613498395>
 1814
 1815 Azzalini D, Rebollo I, Tallon-Baudry C (2019) Visceral signals shape brain dynamics
 1816 and cognition. *Trends in Cognitive Sciences* 23(6):488–509. [https://doi.org/10.1016/](https://doi.org/10.1016/j.tics.2019.03.007)
 1817 [j.tics.2019.03.007](https://doi.org/10.1016/j.tics.2019.03.007)
 1818
 1819 Bahi C, Irrmischer M, Franken K, et al (2024a) Effects of conscious connected breath-
 1820 ing on cortical brain activity, mood and state of consciousness in healthy adults. *Cur-*
 1821 *rent Psychology* 43(12):10578–10589. <https://doi.org/10.1007/s12144-023-05119-6>
 1822
 1823 Bahi C, Irrmischer M, Franken K, et al (2024b) Effects of conscious connected breath-
 1824 ing on cortical brain activity, mood and state of consciousness in healthy adults. *Cur-*
 1825 *rent Psychology* 43(12):10578–10589. <https://doi.org/10.1007/s12144-023-05119-6>
 1826
 1827 Balban MY, Neri E, Kogon MM, et al (2023) Brief structured respiration practices
 1828 enhance mood and reduce physiological arousal. *Cell Reports Medicine* 4(1):100895.
 1829 <https://doi.org/10.1016/j.xcrm.2022.100895>
 1830
 1831 Bang H, Ni L, Davis CE (2004) Assessment of blinding in clinical trials. *Controlled*
 1832 *Clinical Trials* 25(2):143–156. <https://doi.org/10.1016/j.cct.2003.10.016>
 1833
 1834 Banushi B, Brendle M, Ragnhildstveit A, et al (2023) Breathwork interventions for
 1835 adults with clinically diagnosed anxiety disorders: A scoping review. *Brain Sciences*
 1836 13(2):256. <https://doi.org/10.3390/brainsci13020256>
 1837
 1838 Barrett FS, Bradstreet MP, Leoutsakos JMS, et al (2016) The Challenging Experience
 1839 Questionnaire: Characterization of challenging experiences with psilocybin mush-
 1840 rooms. *Journal of Psychopharmacology* 30(12):1279–1295. [https://doi.org/10.1177/](https://doi.org/10.1177/0269881116678781)
 1840 [0269881116678781](https://doi.org/10.1177/0269881116678781)

Barton AC, Do M, Sheen J, et al (2024) The restorative and state enhancing potential of abstract fractal-like imagery and interactive mindfulness interventions in virtual reality. *Virtual Reality* 28(1):53

Bartram L, Nakatani A (2009) Distinctive parameters of expressive motion. In: *Computational Aesthetics in Graphics, Visualization, and Imaging*. The Eurographics Association, pp 129–136, <https://doi.org/10.2312/COMPAESTH/COMPAESTH09/129-136>, URL <http://diglib.eg.org/handle/10.2312/COMPAESTH.COMPAESTH09.129-136>

Belli F, Felisatti A, Fischer MH (2021) BreaThink: breathing affects production and perception of quantities. *Experimental Brain Research* 239(8):2489–2499. <https://doi.org/10.1007/s00221-021-06147-z>, URL <https://doi.org/10.1007/s00221-021-06147-z>

Bentley TKG, D’Andrea-Penna G, Rakic M, et al (2023) Breathing practices for stress and anxiety reduction: Conceptual framework of implementation guidelines based on a systematic review of the published literature. *Brain Sciences* 13(12):1612. <https://doi.org/10.3390/brainsci13121612>

Bertamini M, Makin ADJ, Rampone G (2013) Implicit association of symmetry with positive valence, high arousal and simplicity. *i-Perception* 4(5):317–327. <https://doi.org/10.1068/i0601jw>

Betka S, Canzoneri E, Adler D, et al (2020) Mechanisms of the breathing contribution to bodily self-consciousness in healthy humans: Lessons from machine-assisted breathing? *Psychophysiology* 57(8):e13564. <https://doi.org/10.1111/psyp.13564>, URL <https://doi.org/10.1111/psyp.13564>

Billock VA, Tsou BH (2012) Elementary visual hallucinations and their relationships to neural pattern-forming mechanisms. *Psychological Bulletin* 138(4):744–774. <https://doi.org/10.1037/a0027580>, URL <https://doi.org/10.1037/a0027580>

Blazhenkova O, Kumar MM (2018) Angular versus curved shapes: Correspondences and emotional processing. *Perception* 47(1):67–89. <https://doi.org/10.1177/0301006617731048>, URL <https://doi.org/10.1177/0301006617731048>

Blum J, Rockstroh C, Göritz AS (2019) Heart rate variability biofeedback based on slow-paced breathing with immersive virtual reality nature scenery. *Frontiers in psychology* 10:2172

Blum J, Rockstroh C, Göritz AS (2020) Development and pilot test of a virtual reality respiratory biofeedback approach. *Applied Psychophysiology and Biofeedback* 45(3):153–163

Boiten FA, Frijda NH, Wientjes CJE (1994) Emotions and respiratory patterns: review and critical analysis. *International Journal of Psychophysiology* 17(2):103–128.

1887 [https://doi.org/10.1016/0167-8760\(94\)90027-2](https://doi.org/10.1016/0167-8760(94)90027-2)
1888
1889 Bossenbroek R, Wols A, Weerdmeester J, et al (2020) Efficacy of a virtual reality
1890 biofeedback game (DEEP) to reduce anxiety and disruptive classroom behavior:
1891 single-case study. *JMIR Mental Health* 7(3):e16066. <https://doi.org/10.2196/16066>,
1892 URL <https://doi.org/10.2196/16066>
1893
1894 Bufacchi RJ, Iannetti GD (2018) An action field theory of peripersonal space. *Trends*
1895 *in Cognitive Sciences* 22(12):1076–1090. <https://doi.org/10.1016/j.tics.2018.09.004>
1896
1897 Burger B, Toiviainen P (2020) See how it feels to move: Relationships between
1898 movement characteristics and perception of emotions in dance. *Human Technol-*
1899 *ogy* 16(3):233–256. <https://doi.org/10.17011/ht/urn.202011256764>, URL <https://doi.org/10.17011/ht/urn.202011256764>
1900
1901 Chan PYS, Lee LY, Davenport PW (2024) Neural mechanisms of respiratory intero-
1902 ception. *Autonomic Neuroscience: Basic and Clinical* 253:103181. <https://doi.org/10.1016/j.autneu.2024.103181>
1903
1904 Chang E, Kim HT, Yoo B (2020) Virtual reality sickness: a review of causes and
1905 measurements. *International Journal of Human–Computer Interaction* 36(17):1658–
1906 1682. <https://doi.org/10.1080/10447318.2020.1778351>
1907
1908 Cheiran JFP, Bandeira DR, Pimenta MS (2025) Measuring the key components of the
1909 user experience in immersive virtual reality environments. *Frontiers in Virtual Real-*
1910 *ity* 6:1585614. <https://doi.org/10.3389/frvir.2025.1585614>, URL <https://doi.org/10.3389/frvir.2025.1585614>
1911
1912 Cortez-Vázquez G, Adriaanse M, Burchell GL, et al (2024) Virtual reality breath-
1913 ing interventions for mental health: A systematic review and meta-analysis of
1914 randomized controlled trials. *Applied Psychophysiology and Biofeedback* 49(1):1–
1915 21. <https://doi.org/10.1007/s10484-023-09611-4>, URL <https://doi.org/10.1007/s10484-023-09611-4>
1916
1917 Dambrun M (2016) When the dissolution of perceived body boundaries elicits happi-
1918 ness: The effect of selflessness induced by a body scan meditation. *Consciousness*
1919 *and Cognition* 46:89–98
1920
1921 Dambrun M (2023) Beyond the body/self-consciousness matrix: Scale of the physical
1922 frame of reference and self-transcendence experience. *Psychology of Consciousness: Theory, Research, and Practice* 10(3):292–312. <https://doi.org/10.1037/cns0000314>,
1923 URL <https://doi.org/10.1037/cns0000314>
1924
1925 Damiano C, Gayen P, Rezanejad M, et al (2023) Anger is red, sadness is blue: Emotion
1926 depictions in abstract visual art by artists and non-artists. *Journal of Vision* 23(4):1.
1927 <https://doi.org/10.1167/jov.23.4.1>, URL <https://doi.org/10.1167/jov.23.4.1>
1928
1929
1930
1931
1932

De Rooij A, Broekens J, Lamers MH (2013) Abstract expressions of affect. International Journal of Synthetic Emotions (IJSE) 4(1):1–31	1933 1934 1935
Dietz G, Jane L. E., Washington P, et al (2017) Human perception of swarm robot motion. In: Proceedings of the 2017 CHI Conference Extended Abstracts on Human Factors in Computing Systems. ACM, pp 2520–2527, https://doi.org/10.1145/3027063.3053220 , URL https://doi.org/10.1145/3027063.3053220	1936 1937 1938 1939
Dittrich A (1998) The standardized psychometric assessment of altered states of consciousness (ascs) in humans. Pharmacopsychiatry 31(S2):80–84. https://doi.org/10.1055/s-2007-979351	1940 1941 1942 1943
van Doorn JB, van den Bergh D, Bohm U, et al (2021) The JASP guidelines for conducting and reporting a bayesian analysis. Psychonomic Bulletin & Review 28(3):813–826. https://doi.org/10.3758/s13423-020-01798-5	1944 1945 1946 1947
Ekman P (1993) Facial expression and emotion. American psychologist 48(4):384	1948 1949
Engelen T, Solcà M, Tallon-Baudry C (2023) Interoceptive rhythms in the brain. Nature Neuroscience 26(10):1670–1684. https://doi.org/10.1038/s41593-023-01425-1	1950 1951 1952 1953
Fejer G (2026) Breathing space: Spatial mapping of breath and cardiac biofeedback for affective state representation and coherence training in viscerality. In: Proceedings of the 1st International Conference on Human-Computer Interaction in the Alps (AlpCHI 2026), https://doi.org/10.1145/3780045.3780061 , added to support manuscript introduction citation; verify publisher metadata before submission	1954 1955 1956 1957 1958 1959
Fejer G, Holzapfel T, Gómez-Emilsson A, et al (2025) Viscerality: A bio-responsive vr system for breath-based interactions and coupled oscillator dynamics to augment altered states of consciousness. In: Mensch und Computer 2025 Workshopband, Gesellschaft für Informatik e.V., https://doi.org/10.18420/muc2025-mci-ws11-174 , URL https://dl.gi.de/handle/20.500.12116/46829	1960 1961 1962 1963 1964
Fincham GW, Strauss C, Montero-Marin J, et al (2023) Effect of breathwork on stress and mental health: A meta-analysis of randomised-controlled trials. Scientific Reports 13(1):432. https://doi.org/10.1038/s41598-022-27247-y	1965 1966 1967 1968
Fincham GW, Epel E, Colasanti A, et al (2024) Effects of brief remote high ventilation breathwork with retention on mental health and wellbeing: a randomised placebo-controlled trial. Scientific Reports 14(1):16893	1969 1970 1971 1972
Fincham GW, Caddy E, Kartar AA, et al (2025) Airways to alteration (a2a): a randomised-controlled trial exploring effects of breathwork on conscious state. Research Square https://doi.org/10.21203/rs.3.rs-7254002/v1	1973 1974 1975 1976 1977 1978

- 1979 Gerritsen RJS, Band GPH (2018) Breath of life: The respiratory vagal stimulation
1980 model of contemplative activity. *Frontiers in Human Neuroscience* 12:397. <https://doi.org/10.3389/fnhum.2018.00397>
1981
1982
1983 Glowacki DR (2024) VR models of death and psychedelics: An aesthetic paradigm
1984 for design beyond day-to-day phenomenology. *Frontiers in Virtual Reality*
1985 4:1286950. <https://doi.org/10.3389/frvir.2023.1286950>, URL [https://doi.org/10.](https://doi.org/10.3389/frvir.2023.1286950)
1986 [3389/frvir.2023.1286950](https://doi.org/10.3389/frvir.2023.1286950)
1987
1988 Glowacki DR, Wonnacott MD, Freire R, et al (2020) Isness: Using multi-person VR
1989 to design peak mystical type experiences comparable to psychedelics. In: *Proceed-*
1990 *ings of the 2020 CHI Conference on Human Factors in Computing Systems*. ACM,
1991 pp 1–14, <https://doi.org/10.1145/3313831.3376649>, URL [https://doi.org/10.1145/](https://doi.org/10.1145/3313831.3376649)
1992 [3313831.3376649](https://doi.org/10.1145/3313831.3376649)
1993
1994 Glowacki DR, Williams RR, Wonnacott MD, et al (2022) Group VR experiences can
1995 produce ego attenuation and connectedness comparable to psychedelics. *Scientific*
1996 *Reports* 12(1):8995. <https://doi.org/10.1038/s41598-022-12637-z>, URL [https://doi.](https://doi.org/10.1038/s41598-022-12637-z)
1997 [org/10.1038/s41598-022-12637-z](https://doi.org/10.1038/s41598-022-12637-z)
1998
1999 Goral O, Wald IY, Maimon A, et al (2024a) Enhancing interoceptive sensibili-
2000 ty through exteroceptive–interoceptive sensory substitution. *Scientific Reports*
2001 14(1):14855. <https://doi.org/10.1038/s41598-024-63231-4>, URL [https://doi.org/10.](https://doi.org/10.1038/s41598-024-63231-4)
2002 [1038/s41598-024-63231-4](https://doi.org/10.1038/s41598-024-63231-4)
2003
2004 Goral O, Wald IY, Maimon A, et al (2024b) Enhancing interoceptive sensibili-
2005 ty through exteroceptive–interoceptive sensory substitution. *Scientific Reports*
2006 14(1):14855. <https://doi.org/10.1038/s41598-024-63231-4>, URL [https://doi.org/10.](https://doi.org/10.1038/s41598-024-63231-4)
2007 [1038/s41598-024-63231-4](https://doi.org/10.1038/s41598-024-63231-4)
2008
2009 Grof S (2019) *Holotropic breathwork: New approach to psychotherapy and self-*
2010 *exploration*. In: *Holotropic Breathwork, Second Edition: A New Approach to*
2011 *Self-Exploration and Therapy*. SUNY Press, Albany, NY
2012
2013 Guy M, Normand JM, Jeunet-Kelway C, et al (2023) The sense of embodi-
2014 ment in virtual reality and its assessment methods. *Frontiers in Virtual Real-*
2015 *ity* 4:1141683. <https://doi.org/10.3389/frvir.2023.1141683>, URL [https://doi.org/10.](https://doi.org/10.3389/frvir.2023.1141683)
2016 [3389/frvir.2023.1141683](https://doi.org/10.3389/frvir.2023.1141683)
2017
2018 Hanley AW, Garland EL (2019) Spatial frame of reference as a phenomenological
2019 feature of self-transcendence: Measurement and manipulation through mindfulness
2020 meditation. *Psychology of Consciousness: Theory, Research, and Practice* 6(4):329–
2021 345. <https://doi.org/10.1037/cns0000204>
2022
2023 Hanley AW, Dambrun M, Garland EL (2020) Effects of mindfulness meditation on
2024 self-transcendent states: Perceived body boundaries and spatial frames of reference. *Mindfulness* 11(5):1194–1203

Havenith MN, Leidenberger M, Brasanac J, et al (2025a) Decreased co2 saturation during circular breathwork supports emergence of altered states of consciousness. *Communications Psychology* 3(1):59. <https://doi.org/10.1038/s44271-025-00247-0>

Havenith MN, Leidenberger M, Brasanac J, et al (2025b) Decreased co2 saturation during circular breathwork supports emergence of altered states of consciousness. *Communications Psychology* 3(1). <https://doi.org/10.1038/s44271-025-00247-0>, URL <https://doi.org/10.1038/s44271-025-00247-0>

Homma I, Masaoka Y (2008) Breathing rhythms and emotions. *Experimental Physiology* 93(9):1011–1021. <https://doi.org/10.1113/expphysiol.2008.042424>

Hovmand OR, Madsen M, Fisher PM, et al (2024) Altered states of consciousness in danish healthy volunteers and recreational users of psilocybin and the possible impact of setting and intention: Danish validation of the five-dimensional altered states of consciousness questionnaire. *Journal of Psychopharmacology* 38(9):924–932. <https://doi.org/10.1177/02698811241269669>

Hruby H, Schmidt S, Feinstein JS, et al (2024) Induction of altered states of consciousness during Floatation-REST is associated with the dissolution of body boundaries and the distortion of subjective time. *Scientific Reports* 14(1):9316. <https://doi.org/10.1038/s41598-024-59642-y>, URL <https://doi.org/10.1038/s41598-024-59642-y>

Jachs B (2021) The neurophenomenology of meditative states: Introducing temporal experience tracing to capture subjective experience states and their neural correlates. PhD thesis, University of Cambridge, <https://doi.org/10.17863/CAM.80153>, URL <https://www.repository.cam.ac.uk/handle/1810/332709>

Jerath R, Beveridge C (2020) Respiratory rhythm, autonomic modulation, and the spectrum of emotions: The future of emotion recognition and modulation. *Frontiers in Psychology* 11:1980. <https://doi.org/10.3389/fpsyg.2020.01980>

Johnson ME (2023) Qualia formalism and a symmetry theory of valence. *OpenTheory*, URL https://opentheory.net/Qualia_Formalism_and_a_Symmetry_Theory_of_Valence.pdf

Kaduk J, Cavdan M, Drewing K, et al (2024) From one to many: How active robot swarm sizes influence human cognitive processes. In: 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN). IEEE, pp 1207–1212, <https://doi.org/10.1109/RO-MAN60168.2024.10731232>, URL <https://doi.org/10.1109/RO-MAN60168.2024.10731232>

Karakoc C, Lucifora C, Massimino S, et al (2025) Extending peri-personal space in immersive virtual reality: A systematic review. *Virtual Worlds* 4(1):5. <https://doi.org/10.3390/virtualworlds4010005>, URL <https://doi.org/10.3390/virtualworlds4010005>

2071 Kartar AA, Horinouchi T, Örsik B, et al (2025) Neurobiological substrates of altered
2072 states of consciousness induced by high ventilation breathwork accompanied by
2073 music. PLOS One 20(8):e0329411. <https://doi.org/10.1371/journal.pone.0329411>,
2074 URL <https://doi.org/10.1371/journal.pone.0329411>
2075
2076 Kaufeld M, Bourdeinik J, Prinz LM, et al (2022) Emotions are associated with the
2077 genesis of visually induced motion sickness in virtual reality. Experimental Brain
2078 Research 240(10):2757–2771. <https://doi.org/10.1007/s00221-022-06454-z>
2079
2080 Khatin-Zadeh O, Eskandari Z, Vahdat S, et al (2019) Why are motions effective in
2081 describing emotions? Polish Psychological Bulletin 50(2):119–124. <https://doi.org/10.24425/ppb.2019.126027>
2082
2083 Klüver H (1966) Mescal, and Mechanisms of Hallucinations. University of Chicago
2084 Press, Chicago, reprint edition; originally published 1928
2085
2086 Kodithuwakku Arachchige SNK, Chander H, Freeman H, et al (2022) The walls are
2087 closing in: postural responses to a virtual reality claustrophobic simulation. Clinical
2088 and Translational Neuroscience 6(2):15. <https://doi.org/10.3390/ctn6020015>
2089
2090 Kolahi J, Bang H, Park J (2009) Towards a proposal for assessment of blinding success
2091 in clinical trials: Up-to-date review. Community Dentistry and Oral Epidemiology
2092 37(6):477–484. <https://doi.org/10.1111/j.1600-0528.2009.00494.x>
2093
2094 Kreibig SD (2010) Autonomic nervous system activity in emotion: A review. Biological
2095 Psychology 84(3):394–421. <https://doi.org/10.1016/j.biopsycho.2010.03.010>
2096
2097 Kruschke JK (2021) Bayesian analysis reporting guidelines. Nature Human Behaviour
2098 5(10):1282–1291. <https://doi.org/10.1038/s41562-021-01177-7>
2099
2100 Kuhn RL (2024) A landscape of consciousness: Toward a taxonomy of explanations
2101 and implications. Progress in biophysics and molecular biology 190:28–169
2102
2103 Kwon J, Iedema A (2022) Body and the senses in spatial experience: The implica-
2104 tions of kinesthetic and synesthetic perceptions for design thinking. Frontiers in
2105 Psychology 13:864009. <https://doi.org/10.3389/fpsyg.2022.864009>
2106
2107 Kytikova O, Novgorodtseva T, Denisenko Y, et al (2021) Thermosensory transient
2108 receptor potential ion channels and asthma. Biomedicines 9(7):816. <https://doi.org/10.3390/biomedicines9070816>
2109
2110 Lawrence DW, Carhart-Harris R, Griffiths R, et al (2022) Phenomenology and content
2111 of the inhaled N, N-dimethyltryptamine (N, N-DMT) experience. Scientific Reports
2112 12(1):8562. <https://doi.org/10.1038/s41598-022-11999-8>, URL <https://doi.org/10.1038/s41598-022-11999-8>
2113
2114
2115
2116

Lenggenhager B, Tadi T, Metzinger T, et al (2007) Video ergo sum: Manipulating
bodily self-consciousness. *Science* 317(5841):1096–1099. <https://doi.org/10.1126/science.1143439>, URL <https://doi.org/10.1126/science.1143439>

Lewis-Healey E, Tagliazucchi E, Canales-Johnson A, et al (2024) Breathwork-induced
psychedelic experiences modulate neural dynamics. *Cerebral Cortex* 34(8):bhae347

Makin ADJ, Wilton MM, Pecchinenda A, et al (2012) Symmetry perception and
affective responses: A combined eeg/emg study. *Neuropsychologia* 50(14):3250–3261.
<https://doi.org/10.1016/j.neuropsychologia.2012.10.003>

Makin ADJ, Roccato M, Karakashevskaya E, et al (2023) Symmetry percep-
tion and psychedelic experience. *Symmetry* 15(7):1340. <https://doi.org/10.3390/sym15071340>

Maravita A, Iriki A (2004) Tools for the body (schema). *Trends in Cognitive Sciences*
8(2):79–86. <https://doi.org/10.1016/j.tics.2003.12.008>

Marcolin F, Wally Scurati G, Ulrich L, et al (2021) Affective virtual reality: How to
design artificial experiences impacting human emotions. *IEEE Computer Graphics
and Applications* 41(6):171–178. <https://doi.org/10.1109/MCG.2021.3115015>, URL
<https://doi.org/10.1109/MCG.2021.3115015>

McGraw KO, Wong SP (1992) A common language effect size statistic. *Psychological
Bulletin* 111(2):361–365. <https://doi.org/10.1037/0033-2909.111.2.361>, URL <https://doi.org/10.1037/0033-2909.111.2.361>

Menezes KKP, Nascimento LR, Ada L, et al (2016) Respiratory muscle training
increases respiratory muscle strength and reduces respiratory complications after
stroke: a systematic review. *Journal of Physiotherapy* 62(3):138–144. <https://doi.org/10.1016/j.jphys.2016.05.014>, URL <https://doi.org/10.1016/j.jphys.2016.05.014>

Millière R (2020) The varieties of selflessness. *Philosophy and the Mind Sci-
ences* 1(I):1–41. <https://doi.org/10.33735/phimisci.2020.i.48>, URL <https://doi.org/10.33735/phimisci.2020.i.48>

Mogan R, Fischer R, Bulbulia JA (2017) To be in synchrony or not? a meta-analysis
of synchrony’s effects on behavior, perception, cognition and affect. *Journal of
Experimental Social Psychology* 72:13–20. <https://doi.org/10.1016/j.jesp.2017.03.009>

Monti A, Porciello G, Tieri G, et al (2020a) The “embreathment” illusion high-
lights the role of breathing in corporeal awareness. *Journal of Neurophysiology*
123(1):420–427. <https://doi.org/10.1152/jn.00617.2019>, URL <https://doi.org/10.1152/jn.00617.2019>

2163 Monti A, Porciello G, Tieri G, et al (2020b) The “embreathment” illusion high-
 2164 lights the role of breathing in corporeal awareness. *Journal of Neurophysiology*
 2165 123(1):420–427. <https://doi.org/10.1152/jn.00617.2019>, URL <https://doi.org/10.1152/jn.00617.2019>
 2166
 2167
 2168 Morey RD, Rouder JN (2023) BayesFactor: Computation of Bayes Factors for
 2169 Common Designs. URL <https://github.com/richarddmorey/bayesfactor>, r package
 2170 version 0.9.12-4.7
 2171
 2172 Mori K, Sakano H (2022a) Neural circuitry for stress information of environmental
 2173 and internal odor worlds. *Frontiers in Behavioral Neuroscience* 16:943647. <https://doi.org/10.3389/fnbeh.2022.943647>
 2174
 2175 Mori K, Sakano H (2022b) Processing of odor information during the respiratory cycle
 2176 in mice. *Frontiers in Neural Circuits* 16:861800. <https://doi.org/10.3389/fncir.2022.861800>
 2177
 2178
 2179 Muth C, Carbon CC (2013) The aesthetic aha: On the pleasure of having insights into
 2180 gestalt. *Acta psychologica* 144(1):25–30
 2181
 2182 Pancini E, Di Natale AF, Villani D (2025) Breathing in virtual reality for promoting
 2183 mental health: a scoping review. *Virtual Reality* 29(1):29. <https://doi.org/10.1007/s10055-024-01096-8>, URL <https://doi.org/10.1007/s10055-024-01096-8>
 2184
 2185
 2186 Patrizz A, El Hamamy A, Maniskas M, et al (2023) Stroke-induced respira-
 2187 tory dysfunction is associated with cognitive decline. *Stroke* 54(7):1863–1874.
 2188 <https://doi.org/10.1161/STROKEAHA.122.041239>, URL <https://doi.org/10.1161/STROKEAHA.122.041239>
 2189
 2190
 2191 Pecchinenda A, Bertamini M, Makin AD, et al (2014) The pleasantness of visual
 2192 symmetry: always, never or sometimes. *PLoS ONE* 9(3):e92685
 2193
 2194 Petkova VI, Ehrsson HH (2008) If I were you: Perceptual illusion of body swapping.
 2195 *PLoS ONE* 3(12):e3832. <https://doi.org/10.1371/journal.pone.0003832>, URL <https://doi.org/10.1371/journal.pone.0003832>
 2196
 2197
 2198 Petrizzo I, Mikellidou K, Avraam S, et al (2024) Reshaping the periper-
 2199 sonal space in virtual reality. *Scientific Reports* 14(1). <https://doi.org/10.1038/s41598-024-52383-y>, URL <https://doi.org/10.1038/s41598-024-52383-y>
 2200
 2201 Pöhlmann KMT, Föcker J, Dickinson P, et al (2021) The effect of motion direction
 2202 and eccentricity on vection, VR sickness and head movements in virtual reality.
 2203 *Multisensory Research* 34(6):623–662. <https://doi.org/10.1163/22134808-bja10049>
 2204
 2205 Pollick FE, Paterson HM, Bruderlin A, et al (2001) Perceiving affect from arm move-
 2206 ment. *Cognition* 82(2):B51–B61. [https://doi.org/10.1016/S0010-0277\(01\)00147-0](https://doi.org/10.1016/S0010-0277(01)00147-0),
 2207 URL [https://doi.org/10.1016/S0010-0277\(01\)00147-0](https://doi.org/10.1016/S0010-0277(01)00147-0)
 2208

Prpa M, Fdili-Alaoui S, Schiphorst T, et al (2020) Articulating experience. In: Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems. ACM, pp 1–14, https://doi.org/10.1145/3313831.3376664 , URL https://doi.org/10.1145/3313831.3376664	2209 2210 2211 2212 2213
Prugger J, Derdiyok E, Dinkelacker J, et al (2022) The altered states database: Psychometric data from a systematic literature review. <i>Scientific Data</i> 9(1):720. https://doi.org/10.1038/s41597-022-01822-4	2214 2215 2216 2217
Qin Z, Yu Y, Gu H, et al (2024) Sas macro programme for Bang’s Blinding Index to assess and visualise the success of blinding in randomised controlled trials. <i>General Psychiatry</i> 37:e101578. https://doi.org/10.1136/gpsych-2024-101578	2218 2219 2220 2221
Ramachandran VS, Hirstein W (1999) The science of art: A neurological theory of aesthetic experience. <i>Journal of Consciousness Studies</i> 6(6–7):15–31	2222 2223 2224
Reeves WT (1983) Particle systems—a technique for modeling a class of fuzzy objects. <i>ACM Transactions on Graphics</i> 2(2):91–108. https://doi.org/10.1145/357318.357320 , URL https://doi.org/10.1145/357318.357320	2225 2226 2227
Reynolds CW (1987) Flocks, herds, and schools: A distributed behavioral model. In: Proceedings of the 14th Annual Conference on Computer Graphics and Interactive Techniques. ACM, pp 25–34, https://doi.org/10.1145/37401.37406 , URL https://doi.org/10.1145/37401.37406	2228 2229 2230 2231 2232
Rhinewine JP, Williams OJ (2007) Holotropic breathwork: The potential role of a prolonged, voluntary hyperventilation procedure as an adjunct to psychotherapy. <i>Journal of Alternative and Complementary Medicine</i> 13(7):771–776. https://doi.org/10.1089/acm.2006.6203	2233 2234 2235 2236 2237
Rockstroh C, Blum J, Göritz AS (2019) Virtual reality in the application of heart rate variability biofeedback. <i>International Journal of Human-Computer Studies</i> 130:209–220. https://doi.org/10.1016/j.ijhcs.2019.06.011 , URL https://doi.org/10.1016/j.ijhcs.2019.06.011	2238 2239 2240 2241 2242
Rockstroh C, Blum J, Göritz AS (2021) A mobile VR-based respiratory biofeedback game to foster diaphragmatic breathing. <i>Virtual Reality</i> 25(2):539–552. https://doi.org/10.1007/s10055-020-00471-5 , URL https://doi.org/10.1007/s10055-020-00471-5	2243 2244 2245 2246 2247
Rodrigues FA, Peron TKDM, Ji P, et al (2016) The kuramoto model in complex networks. <i>Physics Reports</i> 610:1–98. https://doi.org/10.1016/j.physrep.2015.10.008	2248 2249 2250
Rogers B, Graham M (1979) Motion parallax as an independent cue for depth perception. <i>Perception</i> 8(2):125–134. https://doi.org/10.1068/p080125 , URL https://doi.org/10.1068/p080125	2251 2252 2253 2254

2255 Romeo B, Kervadec E, Fauvel B, et al (2025) The intensity of the psychedelic experience is reliably associated with clinical improvements: A systematic review and
 2256 meta-analysis. *Neuroscience & Biobehavioral Reviews* 172:106086. <https://doi.org/10.1016/j.neubiorev.2025.106086>, URL <https://doi.org/10.1016/j.neubiorev.2025.106086>
 2257
 2258
 2259
 2260
 2261 de Rooij A, Broekens J, Lamers MH (2013) Abstract expressions of affect. *International Journal of Synthetic Emotions* 4(1):1–31. <https://doi.org/10.4018/jse.2013010101>, URL <https://doi.org/10.4018/jse.2013010101>
 2262
 2263
 2264
 2265 van Rooij M, Lobel A, Harris O, et al (2016) DEEP: A biofeedback virtual reality game for children at-risk for anxiety. In: *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. ACM, pp 1989–1997, <https://doi.org/10.1145/2851581.2892452>, URL <https://doi.org/10.1145/2851581.2892452>
 2266
 2267
 2268
 2269 Rook C, Panda S, Sice P, et al (2026) Heart garden: Visualizing heart rate data using the metaphor of a garden. *International Journal of Human–Computer Interaction* 42(1):114–130. <https://doi.org/10.1080/10447318.2025.2504201>, URL <https://doi.org/10.1080/10447318.2025.2504201>
 2270
 2271
 2272
 2273
 2274 Roseman L, Nutt DJ, Carhart-Harris RL (2018) Quality of acute psychedelic experience predicts therapeutic efficacy of psilocybin for treatment-resistant depression. *Frontiers in Pharmacology* 8:974. <https://doi.org/10.3389/fphar.2017.00974>
 2275
 2276
 2277
 2278 Rubin DS, Morgan RC, Pinchbeck D (2010) *Psychedelic: Optical and Visionary Art Since the 1960s*. MIT Press, <https://doi.org/10.5860/choice.48-0074>
 2279
 2280
 2281 Ruta N, Mastandrea S, Penacchio O, et al (2021) When symmetric and curved visual contour meet intentional instructions: Hedonic value and preference. *Quarterly Journal of Experimental Psychology* 74(9):1525–1541. <https://doi.org/10.1177/17470218211007388>
 2282
 2283
 2284
 2285
 2286 Sabnis AS, Shadid M, Yost GS, et al (2008) Human lung epithelial cells express a functional cold-sensing TRPM8 variant. *American Journal of Respiratory Cell and Molecular Biology* 39(4):466–474. <https://doi.org/10.1165/rcmb.2007-0440OC>
 2287
 2288
 2289
 2290 Sanders JW, Zijlmans J (2021) Moving past mysticism in psychedelic science. *ACS Pharmacology & Translational Science* 4(3):1253–1255. <https://doi.org/10.1021/acsptsci.1c00097>, URL <https://doi.org/10.1021/acsptsci.1c00097>
 2291
 2292
 2293
 2294 Santos M, Egerstedt M (2021) From motions to emotions: Can the fundamental emotions be expressed in a robot swarm? *International Journal of Social Robotics* 13(4):751–764. <https://doi.org/10.1007/s12369-020-00665-6>, URL <https://doi.org/10.1007/s12369-020-00665-6>
 2295
 2296
 2297
 2298 Sarkar AA (2017) Functional correlation between breathing and emotional states. *MOJ Anatomy & Physiology* 3(5):157–158. [https://doi.org/10.15406/mojap.2017.3\(5\).157-158](https://doi.org/10.15406/mojap.2017.3(5).157-158)
 2299
 2300

03.00108	2301
	2302
Seiler A, Camilo M, Korostovtseva L, et al (2019) Prevalence of sleep-	2303
disordered breathing after stroke and TIA. <i>Neurology</i> 92(7):e648–e654. https://doi.org/10.1212/WNL.0000000000006904 , URL https://doi.org/10.1212/WNL.0000000000006904	2304
	2305
	2306
	2307
Serino A (2019) Peripersonal space (PPS) as a multisensory interface between	2308
the individual and the environment, defining the space of the self. <i>Neuroscience</i>	2309
& <i>Biobehavioral Reviews</i> 99:138–159. https://doi.org/10.1016/j.neubiorev.2019.01.016	2310
	2311
	2312
Shamekhi A, Bickmore T (2018) Breathe deep. In: <i>Proceedings of the 12th EAI Inter-</i>	2313
<i>national Conference on Pervasive Computing Technologies for Healthcare</i> . ACM, pp	2314
108–117, https://doi.org/10.1145/3240925.3240940 , URL https://doi.org/10.1145/3240925.3240940	2315
	2316
	2317
Shi L, Liu F, Liu Y, et al (2023) Biofeedback respiratory rehabilitation training system	2318
based on virtual reality technology. <i>Sensors</i> 23(22):9025. https://doi.org/10.3390/s23229025 , URL https://doi.org/10.3390/s23229025	2319
	2320
	2321
Siebieszuk A, Płoński AF, Baranowski M (2025) Breathwork for chronic stress	2322
and mental health: Does choosing a specific technique matter? <i>Medical Sciences</i>	2323
13(3):127. https://doi.org/10.3390/medsci13030127	2324
	2325
Spee BT, Arato J, Mikuni J, et al (2024) The dynamics of experiencing gestalt and	2326
aha in cubist art: pupil responses and art evaluations show a complex interplay of	2327
task, stimuli content, and time course. <i>Frontiers in Psychology</i> 15:1192565	2328
	2329
Spehar B, van Tonder GJ (2017) Koffka’s aesthetic gestalt. <i>Leonardo</i> 50(1):53–57	2330
	2331
Stepanova ER, Desnoyers-Stewart J, Pasquier P, et al (2020) Jel: Breathing together to	2332
connect with others and nature. In: <i>Proceedings of the 2020 ACM Designing Inter-</i>	2333
<i>active Systems Conference</i> . ACM, pp 641–654, https://doi.org/10.1145/3357236.3395532 , URL https://doi.org/10.1145/3357236.3395532	2334
	2335
Stocker K, Hartmann M, Schmid Y, et al (2025) The 3d-ascr scale: A revalidation of	2336
the core dimensions of the altered states of consciousness rating scale 5d(11)-asc	2337
for psychedelic research. <i>Journal of Psychopharmacology</i> https://doi.org/10.1177/02698811251397328	2338
	2339
	2340
Studerus E, Gamma A, Vollenweider FX (2010) Psychometric evaluation of the altered	2341
states of consciousness rating scale (oav). <i>PLOS ONE</i> 5(8):e12412. https://doi.org/10.1371/journal.pone.0012412	2342
	2343
	2344
Suzuki K, Garfinkel SN, Critchley HD, et al (2013) Multisensory integration	2345
across exteroceptive and interoceptive domains modulates self-experience in the	2346

2347 rubber-hand illusion. *Neuropsychologia* 51(13):2909–2917. [https://doi.org/10.1016/](https://doi.org/10.1016/j.neuropsychologia.2013.08.014)
2348 [j.neuropsychologia.2013.08.014](https://doi.org/10.1016/j.neuropsychologia.2013.08.014)
2349
2350 Taves A (2020) Mystical and other alterations in sense of self: An expanded frame-
2351 work for studying nonordinary experiences. *Perspectives on Psychological Science*
2352 15(3):669–690. <https://doi.org/10.1177/1745691619895047>, URL [https://doi.org/](https://doi.org/10.1177/1745691619895047)
2353 [10.1177/1745691619895047](https://doi.org/10.1177/1745691619895047)
2354
2355 Terhune DB, Luke DP, Kaelen M, et al (2016) A placebo-controlled investigation of
2356 synaesthesia-like experiences under LSD. *Neuropsychologia* 88:28–34. [https://doi.](https://doi.org/10.1016/j.neuropsychologia.2016.04.005)
2357 [org/10.1016/j.neuropsychologia.2016.04.005](https://doi.org/10.1016/j.neuropsychologia.2016.04.005)
2358
2359 Tinga AM, Nyklíček I, Jansen MP, et al (2019) Respiratory biofeedback does not facil-
2360 itate lowering arousal in meditation through virtual reality. *Applied Psychophysiol-*
2361 *ogy and Biofeedback* 44(1):51–59. <https://doi.org/10.1007/s10484-018-9421-5>, URL
2362 <https://doi.org/10.1007/s10484-018-9421-5>
2363
2364 Tsakiris M, Tajadura-Jiménez A, Costantini M (2011a) Just a heartbeat away from
2365 one’s body: Interoceptive sensitivity predicts malleability of body-representations.
2366 *Proceedings of the Royal Society B: Biological Sciences* 278(1717):2470–2476. [https:](https://doi.org/10.1098/rspb.2010.2547)
2367 [//doi.org/10.1098/rspb.2010.2547](https://doi.org/10.1098/rspb.2010.2547)
2368
2369 Tsakiris M, Tajadura-Jiménez A, Costantini M (2011b) Just a heartbeat away from
2370 one’s body: Interoceptive sensitivity predicts malleability of body-representations.
2371 *Proceedings of the Royal Society B: Biological Sciences* 278(1717):2470–2476. [https:](https://doi.org/10.1098/rspb.2010.2547)
2372 [//doi.org/10.1098/rspb.2010.2547](https://doi.org/10.1098/rspb.2010.2547)
2373
2374 Uthaug MV, Mason NL, Havenith MN, et al (2021) An experience with Holotropic
2375 Breathwork is associated with improvement in non-judgement and satisfaction with
2376 life while reducing symptoms of stress in a Czech-speaking population. *Journal of*
2377 *Psychedelic Studies* 5(3):176–189. <https://doi.org/10.1556/2054.2021.00193>, URL
2378 <https://doi.org/10.1556/2054.2021.00193>
2379
2380 Vartanian O, Navarrete G, Chatterjee A, et al (2013) Impact of contour on aesthetic
2381 judgments and approach-avoidance decisions in architecture. *Proceedings of the*
2382 *National Academy of Sciences* 110(Supplement 2):10446–10453. [https://doi.org/10.](https://doi.org/10.1073/pnas.1301227110)
2383 [1073/pnas.1301227110](https://doi.org/10.1073/pnas.1301227110)
2384
2385 Vartanian O, Navarrete G, Palumbo L, et al (2024) Neural dissociation between
2386 computational and perceived measures of curvature. *Scientific Reports* 14:8932.
2387 <https://doi.org/10.1038/s41598-024-59525-4>
2388
2389 Vessel EA, Rubin N (2021) From visual perception to aesthetic appeal: Brain
2390 responses to aesthetically appealing natural landscape movies. *Frontiers in Human*
2391 *Neuroscience* 14:612639. <https://doi.org/10.3389/fnhum.2020.612639>
2392

Vicary S, Sperling M, von Zimmermann J, et al (2017) Joint action aesthetics. PLOS ONE 12(7):e0180101. https://doi.org/10.1371/journal.pone.0180101	2393 2394 2395
Vidal J, Andreu CI, Wrzesien M, et al (2024) Inducing selflessness through a numadelic virtual reality experience: A preliminary study. SSRN https://doi.org/10.2139/ssrn.4947164	2396 2397 2398 2399
Vidal J, Andreu CI, Wrzesien M, et al (2025) Inducing selflessness through a numadelic virtual reality experience: a preliminary study. Virtual Reality 30(1). https://doi.org/10.1007/s10055-025-01293-z , URL https://doi.org/10.1007/s10055-025-01293-z	2400 2401 2402 2403
Wagemans J, Elder JH, Kubovy M, et al (2012) A century of Gestalt psychology in visual perception: I. perceptual grouping and figure-ground organization. Psychological Bulletin 138(6):1172–1217. https://doi.org/10.1037/a0029333 , URL https://doi.org/10.1037/a0029333	2404 2405 2406 2407 2408
Wald IY, Maimon A, Keniger de Andrade Gensas L, et al (2023) Breathing based immersive interactions for enhanced agency and body awareness: A claustrophobia motivated study. In: Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, pp 1–7, https://doi.org/10.1145/3544549.3585897 , URL https://doi.org/10.1145/3544549.3585897	2409 2410 2411 2412 2413 2414
Weng HY, Feldman JL, Leggio L, et al (2021) Interventions and manipulations of interoception. Trends in Neurosciences 44(1):52–62. https://doi.org/10.1016/j.tins.2020.09.010	2415 2416 2417 2418
Yoon JM, Im SC, Kim K (2022) Effects of diaphragmatic breathing and pursed lip breathing exercises on the pulmonary function and walking endurance in patients with chronic stroke: a randomised controlled trial. International Journal of Therapy and Rehabilitation 29(8):1–11. https://doi.org/10.12968/ijtr.2021.0027 , URL https://doi.org/10.12968/ijtr.2021.0027	2419 2420 2421 2422 2423 2424
Zaccaro A, Piarulli A, Laurino M, et al (2018) How breath-control can change your life: a systematic review on psycho-physiological correlates of slow breathing. Frontiers in Human Neuroscience 12:353. https://doi.org/10.3389/fnhum.2018.00353	2425 2426 2427 2428
Zaccaro A, Piarulli A, Melosini L, et al (2022) Neural correlates of non-ordinary states of consciousness in pranayama practitioners: the role of slow nasal breathing. Frontiers in Systems Neuroscience 16:803904. https://doi.org/10.3389/fnsys.2022.803904	2429 2430 2431 2432
Zhou Y, Sun B, Li Q, et al (2011) Sensitivity of bronchopulmonary receptors to cold and heat mediated by transient receptor potential cation channel subtypes in an ex vivo rat lung preparation. Respiratory Physiology & Neurobiology 177(3):327–332. https://doi.org/10.1016/j.resp.2011.05.011	2433 2434 2435 2436 2437 2438